

PAST-IoT: Self-supervised IoT intrusion detection via pattern-aware anomaly injection and spectral-temporal dual-tower decomposition

Jiangtao Zhai ^{a,b,*}, Lejun Shen ^a, Shuo Wu ^a, Kai Chen ^a, Guangjie Liu ^{a,b}

^a School of Electronics and Information Engineering, Nanjing University of Information Science and Technology, Nanjing, 210044, China

^b Key Laboratory of Intelligent Support Technology for Complex Environments, Ministry of Education, Nanjing, 210044, China

ARTICLE INFO

Keywords:

Internet of things
Intrusion detection
Self-supervised learning
Anomaly injection
Multi-task learning

ABSTRACT

The large-scale heterogeneous deployment of Internet of Things (IoT) devices exposes limitations in existing Intrusion Detection Systems (IDS), which are severely hindered by labeled data scarcity, highly dynamic attack patterns, and imprecise feature representation. To address these limitations, this paper proposes a novel spectral-temporal decoupling representation learning framework (PAST-IoT). The core idea of PAST-IoT is to represent traffic as a coupling of continuous physical statistics and discrete protocol semantics. First, we construct traffic as multi-channel time series and generate diverse negative samples through a pattern-aware anomaly injection strategy to improve the generalizability of the model to real attacks. Second, a decoupled spectral-temporal dual-tower encoder is designed. The continuous tower adopts Legendre polynomials for global trend extraction. It uses complex-valued self-attention and a gated spectral enhancement module. Thus, it accurately captures amplitude-phase coupling characteristics of traffic fluctuations in the frequency domain. The discrete tower captures the semantic interactions of protocol flags via a context encoder. This design of decoupled representation enables the model to accurately identify covert attacks and protocol violations. Finally, an uncertainty-weighted multi-task loss function is designed to adaptively balance the anomaly contrast, phase-aware reconstruction, auxiliary classification, and masked phase prediction (MPP) tasks. Experiments on multiple datasets demonstrate that the proposed framework significantly outperforms existing models in both anomaly detection and attack classification tasks, with average improvements of 8.87% in accuracy and 14.56% in F1-score, respectively.

1. Introduction

With the rapid development of IoT technology, heterogeneous devices have been widely deployed in critical fields such as smart cities and industrial automation (Adil et al., 2024; Ruan et al., 2023). However, IoT devices are characterized by limited computing resources, diverse communication protocols, and long-term exposure to open networks, making them prime targets for cyberattacks (Cherfi et al., 2026; Li et al., 2022; Wang et al., 2026). As the first line of defense, IDS play an indispensable role (Thakkar & Lohiya, 2023). Traditional rule-based detection methods struggle to cope with highly disguised and variable attacks. Although deep learning-based approaches excel in feature extraction, they still face severe challenges in real-world deployment (Ullah et al., 2024; Zhan et al., 2025).

The extreme scarcity of high-quality labeled data and the shortage of attack samples constitute the main bottleneck that restricts the practical deployment of deep learning models (Gao et al., 2025; Lin & Chen, 2026). In real IoT networks, abnormal traffic is usually submerged in

massive benign background traffic, and new attack patterns emerge continuously. Acquiring accurately labeled attack samples is extremely costly and time consuming (Abdel Wahab, 2022; Jadhav et al., 2026). To alleviate the dependence on manual labels, self-supervised learning (SSL) has emerged as a promising paradigm (Zhao et al., 2024). Nevertheless, most existing SSL frameworks originate from the computer vision (CV) domain (Huang et al., 2023; Shen et al., 2025; Towhid & Shahriar, 2022), and their core data augmentation techniques (e.g. random cropping, Gaussian noise) often destroy the temporal dependence or protocol semantics of network traffic when directly applied. For instance, simply adding random noise to inter-arrival time (IAT) can hardly simulate real network congestion or delay, but instead introduces meaningless disturbances. Random masking may lead to the loss of critical protocol handshake information. Such domain mismatch prevents the model from learning effective features that can characterize real attack logic during the pre-training phase.

The traffic of the IoT network is essentially a complex coupling of continuous physical quantities and discrete protocol states (Altaf et al.,

* Corresponding author.

E-mail addresses: jiangtaozhai@nuist.edu.cn (J. Zhai), 202412180606@nuist.edu.cn (L. Shen), 202412180617@nuist.edu.cn (S. Wu), chenkai@nuist.edu.cn (K. Chen), gjliu@nuist.edu.cn (G. Liu).

<https://doi.org/10.1016/j.eswa.2026.132993>

Received 16 March 2026; Received in revised form 28 April 2026; Accepted 22 May 2026

Available online 26 May 2026

0957-4174/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

2023; Luo et al., 2025; Zha et al., 2026). Existing methods usually directly concatenate these heterogeneous features and map them into real-valued space, ignoring their essential differences in physical meaning. More seriously, many covert low-rate attacks or periodic signaling storms appear stationary in the time domain but exhibit obvious amplitude mutations or phase shifts in the frequency domain (Zhuang et al., 2024). Traditional real-valued models are incapable of capturing phase information, resulting in insufficient detection performance against such complex attacks based on frequency manipulation.

To overcome the limitations of existing methods, this paper proposes a self-supervised intrusion detection framework named PAST-IoT. Unlike conventional approaches that simply concatenate heterogeneous features, the core philosophy of PAST-IoT goes beyond ordinary feature extraction and deeply integrates the physical properties and protocol logic of network traffic. Following this concept, we construct a complete architecture consisting of pattern-aware anomaly injection, spectral-temporal dual-tower encoder, and uncertainty-weighted multi-task pre-training. Detailed technical implementations and mathematical derivations of these modules are comprehensively and elaborately described in Section 3.

The main contributions of this paper are summarized as follows:

1. Unlike the simple homogeneous real-value concatenation of traffic features, this paper proposes a spectral-temporal decoupling representation learning framework for the first time. The framework represents network traffic as a coupling of continuous physical statistics and discrete protocol semantics, and performs separate modeling by constructing orthogonal latent spaces.
2. A complex-domain spectral-temporal dual-tower encoder is designed. By constructing Legendre polynomials for global trend decomposition and introducing a complex-valued self-attention mechanism, this encoder explicitly captures amplitude-phase coupled anomalies and effectively identifies stealthy periodic attacks that evade detection by real-valued models.
3. We construct a collaborative self-supervised driving mechanism. By combining pattern-aware anomaly injection with an uncertainty-weighted multi-task learning strategy, we effectively resolve the optimization conflicts between physical-domain reconstruction and semantic-domain inference, compelling the encoder to converge to a highly discriminative manifold.
4. Extensive experiments on multiple datasets including BoT-IoT and MQTT-IoT-IDS2020 show that PAST-IoT outperforms existing state-of-the-art methods in both detection accuracy and F1-score. Especially in the few-shot scenario with only 1% labeled data, the framework demonstrates superior feature transfer and anti-interference capabilities.

The remaining part of this paper is organized as follows: Section 2 introduces the research status of IoT anomaly detection and attack classification, as well as related self-supervised learning techniques. Section 3 elaborates on the overall architecture of PAST-IoT, including anomaly injection, spectrum-time dual tower encoder, and multi-task pre-training. Section 4 presents the experimental settings and comparison results, and conducts an in-depth analysis. Section 5 summarizes this work.

2. Related works

This section reviews existing research related to IoT intrusion detection. We first analyze the approaches based on supervised learning (SL) and their bottlenecks; next, we discuss the application of self-supervised learning methods in network traffic; finally, we review the relevant advances in frequency-domain representation learning, especially the potential value of phase information in anomaly detection.

2.1. Supervised learning based IoT intrusion detection

Due to its powerful ability to automatically extract features, deep learning technology has been widely applied to IoT intrusion detection

(Luo et al., 2023; Wang et al., 2023; Wu & Wang, 2026). Early studies mainly focused on using Convolutional Neural Networks (CNN) to learn features from spatial local patterns of network traffic, or feeding traffic sequences into Recurrent Neural Networks (RNN) and their variants to capture temporal dependencies. For example, Nandanwar and Katarya (Nandanwar & Katarya, 2024) proposed an adaptive deep learning model that combines CNN with Gated Recurrent Unit (GRU) to analyze network traffic patterns and detect botnet activities. Lin et al. (2021) proposed a spatiotemporal traffic classification method combining CNN and Long Short-Term Memory (LSTM) to identify traffic in Industrial Internet of Things (IIoT) networks. Zhu et al. (2023) integrated a cost penalty matrix to extract spatiotemporal features for robust representation, and adopted an improved loss function to address the problem of data imbalance. In recent years, the Transformer architecture has shown advantages in modeling long-range dependencies due to its global attention mechanism, and has been adopted for traffic classification and anomaly detection tasks. Wang et al. (2023) proposed an intrusion detection model combining ResNet, Transformer, and BiLSTM, and used the edited nearest neighbor method to alleviate data imbalance. Luo et al. (2023) introduced a Transformer-based IoT device identification method, which classifies traffic data from IoT devices into normal and abnormal categories, and uses another Transformer-based model on normal traffic to recognize IoT device types. Nie et al. (2025) constructed packet-wise header-payload representations for multimodal feature extraction and fusion, significantly reducing computational and spatial resource overhead.

However, most existing deep learning-based methods rely heavily on a large amount of labeled data for supervised training. In real-world IoT scenarios, attack samples are scarce and new attacks emerge continuously, leading to severe data imbalance and insufficient generalization ability of the models. Furthermore, most existing methods adopt homogeneous processing strategies, such as simply concatenating or mapping continuous features and discrete features into the same real-valued space. Such processing ignores the inherent nature that continuous variables reflect the physical fluctuations of traffic and discrete variables carry the logical interactions of protocols. This feature confusion makes it difficult for the model to accurately capture fine-grained abnormal patterns under sophisticated attacks.

2.2. Self-supervised learning based IoT intrusion detection

To reduce reliance on labeled data, SSL has achieved tremendous success in CV and natural language processing (NLP) by designing pre-text tasks to learn universal representations from unlabeled data (Zong et al., 2025). This paradigm has also been introduced into network traffic analysis. Common SSL methods include: contrastive learning, which trains models to embed similar samples closer and dissimilar ones further apart by constructing positive and negative sample pairs; and reconstruction learning, which forces models to learn the intrinsic structure and distribution of data through denoising autoencoders or masked reconstruction tasks (Eslami & Hamouda, 2025; Wang et al., 2025; Zhao et al., 2026; Zhong et al., 2026). Li et al. (2022) designed an anomaly detection framework consisting of a traffic capturer, a traffic preprocessor, and an anomaly detector. Each detector is built using an LSTM autoencoder, which eliminates the need for labeled attack data and enables the handling of emerging attacks. Darban et al. (2025) proposed an end-to-end self-supervised representation learning method that redefines positive and negative samples to create distinct representations for normal and abnormal instances. Meidan et al. (2023) employed an autoencoder to distinguish between frequent and infrequent traffic flows, then performed cluster analysis only on infrequent flows to classify them as known or unknown. Tsogbaatar et al. (2021) adopted deep stacked autoencoders for feature extraction, which are then integrated into an ensemble learning model. This framework can detect anomalies and predict the short-term and long-term states of devices. Abusitta et al. (2023) proposed an anomaly detection method based on a denoising

Table 1
Comparison of existing IoT intrusion detection models.

Method	Backbone Network	Traffic Representation	Paradigm	Frequency Domain Usage
Lin et al. (2021)	CNN + Bi-LSTM	Byte sequence	SL	×
Zhu et al. (2023)	CNN + Bi-LSTM	Byte sequence	SL	×
Wang et al. (2023)	ResNet + Transformer + Bi-LSTM	Statistical features	SL	×
Luo et al. (2023)	Transformer	Statistical features	SL	×
Nandanwar and Katarya (2024)	CNN + GRU	Statistical features	SL	×
Nie et al. (2025)	CNN + Multi-Head Self-Attention	Byte sequence	SL	×
Tsogbaatar et al. (2021)	Autoencoder	Statistical features	SSL	×
Li et al. (2022)	LSTM Autoencoder	Byte sequence	SSL	×
Meidan et al. (2023)	Autoencoder + K-means	Statistical features	SSL	×
Abusitta et al. (2023)	Denosing Autoencoder	Statistical features	SSL	×
Xiong et al. (2023)	MLP + CNN	Time series	SL	✓
Dillbary et al. (2024)	Random Forest	Time series	SL	✓
Zhang et al. (2025)	Tree-structured LSTM	Time series	SL	✓
Alsulami and Alturki (2025)	Random Forest + LSTM + GRU	Statistical features	SL	✓
PAST-IoT(our)	Spectral-Temporal Dual-Tower Encoder	Three-channel time series	SSL	✓

autoencoder, combining self-supervised pre-training with fine-tuning to improve the accuracy of malicious data detection. However, transferring SSL to the field of network traffic still has limitations. Network traffic exhibits strict protocol semantics and temporal causal logic; therefore, naive data augmentation or random masking may disrupt its true temporal dynamics or generate meaningless samples that violate protocol specifications. The lack of an anomaly injection mechanism based on protocol standards and the physical characteristics of traffic results in pre-trained models only learning shallow statistical biases, rather than understanding deep attack logic. Therefore, it is necessary to design domain-adaptive pre-training tasks and sample generation strategies to enable models to truly grasp the essence of network behavior.

2.3. Representation learning in frequency domain

Information that is difficult to detect in the time domain may manifest prominently in the frequency domain. Therefore, leveraging frequency-domain information for time series analysis has emerged as a crucial research direction. Traditional methods, such as the Fourier transform and wavelet transform, are employed to extract frequency-domain features from time series, which are then combined with machine learning models for classification. In the field of deep learning, several studies attempt to transform time-domain signals into the frequency domain and apply neural networks therein, or design hybrid architectures that simultaneously process both time-domain and frequency-domain information. Dillbary et al. (2024) utilize time series combined with the Short-Time Fourier Transform (STFT) to extract statistical and spectral features from spectrograms, offering a novel perspective for encrypted traffic classification. Zhang et al. (2025) employ the Discrete Wavelet Transform (DWT) to convert time series into the frequency domain, aiming to discover hidden periodic features. Alsulami and Alturki (2025) propose enhancing the accuracy of network attack prediction by harnessing the powerful synergy between the Continuous Wavelet Transform (CWT) and machine learning ensembles. Xiong et al. (2023) improve the training efficiency of network traffic classification models by pre-analyzing the spectra obtained from the Fourier transform of input samples to estimate required hyperparameters. Liu et al. (2023) treat raw time-series data and their corresponding spectral data as two independent views, using the frequency-domain view to learn invariant frequency patterns in the data and performing contrastive learning with the time-domain view.

However, existing works are primarily limited to amplitude spectrum analysis in the real domain, neglecting the value of phase information in the complex domain. While amplitude determines the energy intensity of a signal, phase encapsulates the structural and time-delay information of the signal waveform. For stealthy low-rate attacks or periodic scans, the amplitude variations of traffic are minimal, and anomalies are primarily reflected in phase shifts or misalignments. Conventional

real-valued neural networks cannot directly process complex-valued inputs, leading to the loss of critical phase-coupled features during feature extraction. To address this, this paper proposes a complex-domain spectral-temporal dual-tower encoder architecture to build a complex-valued neural network capable of simultaneously processing both amplitude and phase information.

In summary, to address the above limitations, this paper proposes a self-supervised framework named PAST-IoT, which includes pattern-aware anomaly injection, spectral-temporal dual-tower encoder, and uncertainty-weighted multi-task pre-training. Table 1 summarizes existing research on IoT intrusion detection.

3. The proposed method

This section elaborates on the proposed PAST-IoT framework. The overall workflow of the framework consists of three core stages: First, a pattern-aware anomaly injection mechanism is adopted to generate diverse negative samples based on the physical characteristics of traffic and protocol specifications. Second, a spectral-temporal dual-tower encoder is constructed, which leverages Legendre polynomials and complex-valued self-attention to decouple traffic features into continuous physical features and discrete semantic features, while capturing fine-grained amplitude-phase coupled features in the frequency domain. Finally, an uncertainty-weighted multi-task learning strategy is employed for pre-training to adaptively balance contrastive learning, phase-aware reconstruction, and masked prediction tasks, thereby learning robust general feature representations for downstream fine-tuning and detection tasks.

3.1. Design motivation and data preprocessing

3.1.1. Design motivation

Network traffic is inherently a coupling of two types of highly heterogeneous data (Liu et al., 2026). Directly concatenating these two distinctly heterogeneous data and mapping them into a single real-valued space not only destroys the discrete semantic integrity, but also obscures the hidden physical frequency-domain laws embedded in continuous data. To address this problem, PAST-IoT adopts the core design concept of heterogeneous feature decoupling. We argue that continuous physical features are essentially wave signals, which should be analyzed in the frequency domain to extract their amplitude and phase characteristics. Meanwhile, discrete protocol features are intrinsically state machines, whose interactive contexts need to be captured via sequence models. PAST-IoT constructs a unified system dually driven by physical understanding and semantic inference, which eliminates mutual interference between heterogeneous features.

3.1.2. Data preprocessing

Based on the above motivations, we transform the original traffic packet capture file (PCAP) into a fixed-length multi-channel sequence. We split a traffic session into several subsequences $\{p_1, p_2, \dots, p_L\}$, where L denotes the sequence length. For each packet p_i , we extract and construct a three-channel feature vector $x_i \in \mathbb{R}^3$. Among these, the continuous physical channel features x_i^{iat} and x_i^{size} are the logarithmic inter-arrival time and the signed log packet size, respectively:

$$x_i^{iat} = \ln(1 + (t_i - t_{i-1}) \times 1000) \quad (1)$$

$$x_i^{size} = \text{dir}_i \cdot \ln(1 + \text{len}(p_i)) \quad (2)$$

In Eq. (1), t_i denotes the timestamp of the current packet, and $\text{dir}_i \in \{+1, -1\}$ indicates the direction of the packet relative to the first packet in the flow. The discrete semantic channel feature x_i^{code} is defined as the protocol status code. For the TCP protocol, we directly extract the decimal value of the Flags field in the TCP header as the status code to preserve its explicit handshake semantics. To address the issue of UDP lacking status flags, we set a sliding window Q_{port} of length 16 that maintains historical destination ports. A dynamic semantic status code is generated by calculating the deviation between the current destination port P_{curr} and the historical ports:

$$x_i^{code} = 256 + \begin{cases} 11, & \text{if } \text{PayloadLen}(p_i) = 0 \\ 7, & \text{if } \min(|P_{curr} - q|) = 0, \forall q \in Q_{port} \\ 8, & \text{if } \min(|P_{curr} - q|) = 1 \\ 9, & \text{if } \min(|P_{curr} - q|) < 10 \\ 12, & \text{if } |\text{Unique}(Q_{port})| > 6 \\ 10, & \text{otherwise} \end{cases} \quad (3)$$

This algorithm isolates UDP behavior from the TCP Flags space using a base offset of 256, enabling the model to directly understand the scanning attack logic under stateless protocols. Other protocols are encoded as 0. Finally, after the above processing, each traffic sample is represented as a tensor $X \in \mathbb{R}^{L \times 3}$.

3.2. Pattern-aware anomaly injection

Existing SSL methods for time series generally adopt Gaussian noise or random masking as data augmentation techniques. While position-preserving masking strategies can effectively maintain sequence structure for discrete categorical data, directly applying standard random masking to the continuous physical features of traffic presents significant drawbacks. Such blind random perturbations introduce artificial high-frequency noise, which distorts the frequency-domain distribution and destroys the continuous amplitude-phase coupling relationship embedded in traffic waveforms. Furthermore, random modifications in the semantic domain readily produce invalid samples that violate strict protocol specifications. To address this issue, we propose a pattern-aware anomaly injector. For a benign sample $X \in \mathbb{R}^{L \times 3}$, where $x_i = [x_i^{iat}, x_i^{size}, x_i^{code}]$, our goal is to generate pseudo-anomalous samples $\tilde{X}$ and their corresponding auxiliary attack-type labels y_{aux} . We design specific injection strategies from two orthogonal dimensions: temporal variation and semantic violation of the protocol, as illustrated in Fig. 1.

The dimension of temporal variation mainly targets the continuous physical features of the channel x_i^{iat} and x_i^{size} , simulating anomalies of the traffic pattern at the network level.

- **Burst:** Simulate the instantaneous high-volume traffic throughput commonly seen in data theft or file transfer attacks. We randomly select a position t in the sequence and apply a significant gain to its packet size to simulate a surge in payload:

$$\tilde{x}_t^{size} = x_t^{size} + \delta \cdot \text{sgn}(x_t^{size}), \quad \delta \sim \mathcal{U}(\delta_{min}, \delta_{max}) \quad (4)$$

In Eq. (4), δ is the burst intensity factor sampled from a uniform distribution. This randomized intensity design is intended to simulate

the fact that in real networks, there are not only attack packets of fixed size, but also diverse payload sizes caused by fragmentation or application-layer differences. The $\text{sgn}(\cdot)$ function is the sign function, which ensures that the original flow direction of the packet remains unchanged when anomalies are injected.

- **DoS Mock:** We compress x_t^{iat} within the sequence to an extremely small interval ϵ , while forcibly setting the packet size to a saturation value Ω to simulate link congestion within a short time:

$$\tilde{x}_t^{iat} = \ln(1 + \epsilon_{dos} \cdot 1000), \quad \epsilon_{dos} \sim \mathcal{U}(\epsilon_{dosmin}, \epsilon_{dosmax}) \quad (5)$$

$$\tilde{x}_t^{size} = \Omega \cdot \text{sgn}(\text{dir}_t), \quad \Omega \sim \mathcal{U}(\Omega_{min}, \Omega_{max}) \quad (6)$$

In Eq. (5), ϵ_{dos} represents the extremely short inter-arrival interval under high-frequency attacks, and Ω denotes the saturation packet size magnitude approaching the maximum transmission unit (MTU). This parameter randomization strategy can simulate the flooding behavior of attacks under different packet rates and payload configurations, preventing the model from overfitting to a single attack frequency.

- **Freeze:** We simulate the interruption of control channels or the cessation of heartbeat in a botnet. For the trailing segment of the sequence, we set x_t^{iat} to an abnormally large value v , and zero out both the packet size and the protocol code to simulate communication silence:

$$\tilde{x}_k^{iat} \sim v, \quad \tilde{x}_k^{size} = 0, \quad \tilde{x}_k^{code} = 0, \quad \forall k \in [t, L] \quad (7)$$

where $v \sim \mathcal{U}(v_{min}, v_{max})$, representing an abnormally long silence duration exceeding the standard TCP/UDP keep-alive timers.

- **Jitter:** We simulate delay jitter caused by network congestion. We inject Gaussian noise $\zeta \sim \mathcal{N}(0, \sigma_j^2)$ only into the x_t^{iat} channel, while keeping the protocol content and packet size unchanged.
- **Noise:** We simulate general network background noise. We add small noise $o \sim \mathcal{N}(0, \sigma_n^2)$ to both x_t^{iat} and x_t^{size} , while leaving the discrete protocol status code unchanged to preserve semantic integrity.
- **Packet Drop:** We simulate network transmission instability or man-in-the-middle interception. We randomly eliminate all feature vectors at selected time steps within the sequence.

The protocol semantic violation dimension mainly focuses on the discrete semantic channels. It injects logical violations based on the TCP/UDP protocol specifications, forcing the model to understand the context interaction of the protocol state.

- **TCP Scan:** We simulate the high-speed probing behavior of port scanning tools. Within a sequence segment, we randomly sample protocol status codes from the scan flag set $\mathcal{F}_{scan}$, and compress x_t^{iat} to an extremely small scan time window to simulate the high-frequency handshake attempts of automated scanning:

$$\tilde{x}_t^{code} \in \mathcal{F}_{scan}, \quad \mathcal{F}_{scan} = \{SYN, RST, SYN-ACK\} \quad (8)$$

$$\tilde{x}_t^{iat} \sim \mathcal{U}(0, \epsilon_{scan}) \quad (9)$$

In Eq. (8), $\mathcal{F}_{scan}$ covers the most prominent handshake initiation and rejection flags in scanning attacks, and ϵ_{scan} is a time threshold approaching zero, reflecting the packet transmission rate of scanners that far exceeds normal human behavior.

- **UDP Injection:** We replace the original protocol code with semantic attack tokens representing specific attack topologies. For the selected injection interval $[t, t + L_{uei}]$, we replace the original protocol code x_k^{code} with a specific attack token c . The attack token set $\mathcal{C}_{attack} = \{c_{seq}, c_{probe}, c_{rand}\}$ corresponds to the following three injection modes:

1) **Sequential Scan Mode (c_{seq}):** Inject an encoding representing a port difference of 1 ($\tilde{x}_k^{code} = 256 + 8$), and forcibly set x_k^{size} to 1 to simulate the attacker's linear traversal of the target host's port range.

2) **Stealth Empty Probe (c_{probe}):** Inject a specific encoding representing zero payload ($\tilde{x}_k^{code} = 256 + 11$), and forcibly set x_k^{size} to 0 to simulate the stealth UDP-Ping operation used for live host discovery.

3) **High-Entropy Random Mode (c_{rand}):** Inject an encoding representing high port entropy ($\tilde{x}_k^{code} = 256 + 12$), and forcibly set x_k^{size}

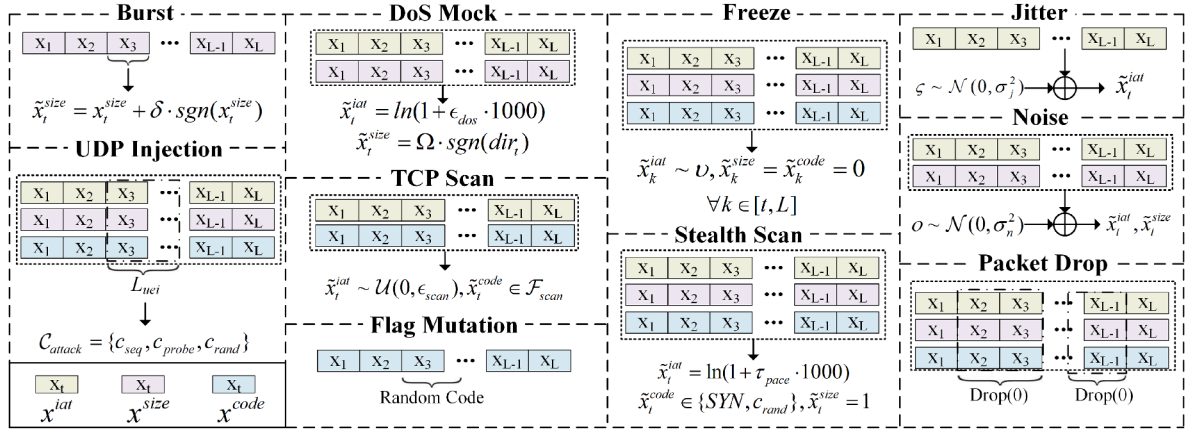


Fig. 1. Pattern-aware anomaly injection.

to 1 to simulate the high-dispersion traffic generated by distributed reflective denial-of-service or random port probing.

Regardless of the injection mode, we apply a unified high-frequency temporal constraint to the injection segment to simulate the high-speed packet transmission characteristics of automated scanners:

$$\tilde{x}_k^{iat} \sim \ln(1 + \tau_{scan} \cdot 1000), \quad \tau_{scan} \sim \mathcal{U}(\tau_{scanmin}, \tau_{scanmax}) \quad (10)$$

In Eq. (10), τ_{scan} is the microsecond-level inter-packet interval, which forces the model to jointly associate discrete attack semantics with high-frequency physical features while learning them. By randomly injecting the above tokens $c \in C_{attack}$, the model is forced to establish a semantic mapping between port behavior patterns and attack intentions in the discrete space, rather than merely memorizing specific port number values.

- **Stealth Scan:** We simulate Low-and-Slow scanning attacks that bypass traffic rate threshold detection. Unlike the bursty characteristics of high-speed scanning, we forcibly set x_t^{iat} to a relatively large value that follows the attack rhythm distribution, set x_t^{size} to 1, and simultaneously mix TCP and UDP probing protocols:

$$\tilde{x}_t^{code} \in \{SYN, c_{rand}\} \quad (11)$$

$$\tilde{x}_t^{iat} = \ln(1 + \tau_{pace} \cdot 1000), \quad \tau_{pace} \sim \mathcal{U}(\tau_{pacemin}, \tau_{pacemax}) \quad (12)$$

In Eq. (12), τ_{pace} represents the pacing delay deliberately introduced by the attacker, sampled from a time interval $[\tau_{pacemin}, \tau_{pacemax}]$ that is significantly higher than the normal flow transmission interval. By sparsely injecting TCP SYN or UDP random port probing packets (c_{rand}) over a long time window, this strategy forces the model to identify stealth scanning behavior by relying solely on the sparse correlation of protocol sequences in the absence of significant statistical burst features.

- **Flag Mutation:** We simulate protocol fuzzing tests or malformed packets. In a local segment, we replace x_t^{code} with randomly generated protocol status codes to train the model to recognize anomalous sequences that do not conform to the standard protocol context.

Notably, the ten injection strategies based on heuristic rules inevitably exhibit a certain distribution bias relative to the complex attack distributions in real-world scenarios. Nevertheless, the fundamental objective of these injection strategies is not to perfectly replicate real attack samples, but to serve as a proxy task for self-supervised contrastive learning. From the perspective of manifold learning, these primitive injection patterns cover orthogonal physical distortions and semantic violations, acting as hard negatives in the latent space. They force the encoder to abandon simple statistical shortcuts and instead construct a highly compact manifold boundary for normal traffic. Consequently, while the pre-training phase relies on predefined rules, it establishes a

Algorithm 1 Pattern-aware anomaly injection.

Require: Benign sample $X \in \mathbb{R}^{L \times 3}$, consisting of x_t^{iat} , x_t^{size} , and x_t^{code}
Ensure: Pseudo-anomalous sample $\tilde{X}$, auxiliary label y_{aux}

- 1: Initialize $\tilde{X} \leftarrow X$
- 2: Select anomaly dimension: Temporal Variation or Protocol Semantic Violation
- 3: **if** Temporal Variation Dimension **then**
 - 4: **Burst** (Eq. 4): $\tilde{x}_t^{size} = x_t^{size} + \delta \cdot \text{sgn}(x_t^{size})$, where $\delta \sim \mathcal{U}(\delta_{min}, \delta_{max})$
 - 5: **DoS Mock** (Eq. 5, 6): $\tilde{x}_t^{iat} = \ln(1 + \epsilon_{dos} \cdot 1000)$, $\tilde{x}_t^{size} = \Omega \cdot \text{sgn}(dir_t)$
 - 6: **Freeze** (Eq. 7): For $k \in [1, L]$, set $\tilde{x}_k^{iat} \sim \mathcal{U}$, $\tilde{x}_k^{size} = 0$, $\tilde{x}_k^{code} = 0$
 - 7: **Jitter & Noise:** Add Gaussian noise to x_t^{iat} and x_t^{size} appropriately
 - 8: **Packet Drop:** Randomly zero out the entire feature vectors at selected time steps
- 9: **else if** Protocol Semantic Violation Dimension **then**
 - 10: **TCP Scan** (Eq. 8, 9): $\tilde{x}_t^{code} \in \mathcal{F}_{scan}$, $\tilde{x}_t^{iat} \sim \mathcal{U}(0, \epsilon_{scan})$
 - 11: **UDP Injection** (Eq. 10): Inject token $c \in \{c_{seq}, c_{probe}, c_{rand}\}$ and set $\tilde{x}_t^{iat} = \ln(1 + \tau_{scan} \cdot 1000)$
 - 12: **Stealth Scan** (Eq. 11, 12): $\tilde{x}_t^{code} \in \{SYN, c_{rand}\}$, $\tilde{x}_t^{iat} = \ln(1 + \tau_{pace} \cdot 1000)$
 - 13: **Flag Mutation:** Replace x_t^{code} with randomly generated protocol status codes
- 14: **end if**
- 15: $y_{aux} \leftarrow$ Assigned pseudo-attack category based on the injected anomaly
- 16: **return** $\tilde{X}, y_{aux}$

robust and generalized feature space. This distinct boundary gives the model strong feature transferability in the fine-tuning phase. With limited labeled data, the model can quickly adapt to complex, real-world attack variants unseen during pre-training. It also accurately identifies these variants. The detailed injection mechanism is shown in Algorithm 1.

3.3. Spectral-temporal dual-tower encoder

The traffic of the IoT network exhibits significant heterogeneity: its physical features show continuous numerical fluctuations that follow time-varying physical laws, while the protocol features display discrete state transitions governed by strict syntactic interaction logic. To address the limitation of traditional real-valued models that indiscriminately process heterogeneous features, we design a spectral-temporal dual-tower encoder. This module consists of two parallel pro-

cessing paths that perform decoupled modeling of traffic in the complex frequency domain and discrete semantic space, respectively. A multi-scale convolution module and a global statistics module are incorporated to improve feature completeness. To ensure the fusibility of dual-tower features in the latent space, we unify the dimension of the feature of both paths to d_{model} . The detailed design is illustrated in Fig. 2.

3.3.1. Continuous tower

The continuous tower aims to capture the fluctuation patterns and fingerprints in the frequency-domain of traffic at the physical level. The input is the continuous feature sequence $X_{cont} \in \mathbb{R}^{L \times 2}$. To handle non-stationarity, we first adopt Reversible Instance Normalization (RevIN) to eliminate distribution shift. Subsequently, deep feature extraction is performed through the following three cascaded modules:

1. **Legendre Polynomial Trend Decomposition:** To separate long-term evolutionary trends and short-term local fluctuations of traffic signals, we adopt a learnable basis expansion mechanism based on Legendre polynomials for trend extraction. Although perfect analytical orthogonality is difficult to maintain for discrete and non-uniformly sampled traffic sequences, employing Legendre polynomials as projection bases provides critical numerical stability. Such orthogonal basis expansion alleviates gradient-related issues and smoothly projects global trends into a low-dimensional coefficient space. For the input sequence X_{cont} , we construct a D th order Legendre polynomial basis matrix $\mathbf{P} \in \mathbb{R}^{L \times (D+1)}$ in the normalized time domain. The global trend term $\mathcal{T}$ is obtained by fitting the linear regression coefficient:

$$\mathcal{T} = \mathbf{P} \cdot \mathbf{W}_{trf}(X_{cont}) \quad (13)$$

In Eq. (13), $\mathbf{W}_{trf}$ is a linear layer to learn polynomial coefficients, and $S = X_{cont} - \mathcal{T}$ is the seasonal component containing high-frequency information, which is then fed into a complex attention module for fine-grained modeling. This design effectively prevents the global trend from masking subtle periodic attack features.

2. **Complex Self-Attention Mechanism:** To capture phase-coupled features hidden in the frequency domain, we map S to the complex domain. It is particularly noteworthy that for low-rate or stealthy attacks, anomalies seldom manifest as large amplitude spikes in the residual components. Instead, their malicious fingerprints are concealed in the periodic phase shifts of the packet transmission rhythm. Accordingly, relying solely on real-valued amplitude analysis of the residuals is insufficient; a mechanism capable of capturing relative phase consistency is strictly required. First, the sequence is split into L/p non-overlapping patches. Two independent linear mapping layers generate the amplitude A and phase ϕ , constructing the complex representation $Z = A \cdot e^{j\phi} \in \mathbb{C}^{(L/p) \times d_{model}}$. In the complex domain, the attention mechanism not only measures the similarity of amplitudes but also needs to perceive the degree of phase alignment. We define complex queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$. The calculation of complex attention scores is based on the real part of the Hermitian inner product. By expressing $\mathbf{Q}$ and $\mathbf{K}$ in polar coordinates with amplitudes $|\mathbf{Q}|, |\mathbf{K}|$ and phases θ_Q, θ_K , the score can be equivalently formulated as:

$$Score(\mathbf{Q}, \mathbf{K}) = Re(\mathbf{Q}\mathbf{K}^H) = \mathbf{Q}_r\mathbf{K}_r^T + \mathbf{Q}_i\mathbf{K}_i^T = |\mathbf{Q}||\mathbf{K}| \cos(\theta_Q - \theta_K) \quad (14)$$

$$Attention(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = Softmax\left(\frac{Score(\mathbf{Q}, \mathbf{K})}{\sqrt{d_k}}\right) \cdot \mathbf{V} \quad (15)$$

In Eq. (14), $\mathbf{K}^H$ denotes the conjugate transpose, and the subscripts r and i represent the real and imaginary parts, respectively. This derivation elegantly proves that although taking only the real part might intuitively seem to discard information, it explicitly encodes the relative phase difference ($\Delta\theta = \theta_Q - \theta_K$). Therefore, if a stealthy attack manipulates the transmission timing (introducing a phase shift) while keeping the traffic volume (amplitude) normal, the $\cos(\Delta\theta)$ term will directly penalize the attention score. Thus, extracting the real part effectively captures the amplitude-phase coupling without losing critical temporal alignment information, enabling the model to precisely identify stealthy

attacks. After complex attention interaction, we obtain the deep complex sequence representation Z_{out} . To extract salient time-domain features and compress the sequence dimension, we perform max-pooling on the real and imaginary parts separately, and concatenate them to obtain the continuous time-domain feature vector $z_{cont} \in \mathbb{R}^{2 \times d_{model}}$.

3. **Gated Spectral Enhancer:** To filter out meaningless high-frequency noise and enhance attack-related spectral features, we introduce a frequency-domain gating mechanism after the attention layer:

$$\tilde{Z}_{freq} = IFFT(FFT(Z_{out}) \odot \mathbf{W}_{freq}) \quad (16)$$

$$Z_{spec} = Re(\tilde{Z}_{freq}) \odot Sigmoid\left(\mathbf{W}_{gate}\left([Re(\tilde{Z}_{freq}), Im(\tilde{Z}_{freq})]\right)\right) \quad (17)$$

In Eq. (16), $FFT(\cdot)$ and $IFFT(\cdot)$ are the Fast Fourier Transform and its inverse, respectively, enabling the switch between time and frequency domains; $\mathbf{W}_{freq}$ and $\mathbf{W}_{gate}$ are learnable complex frequency-domain weights and a linear gating layer, respectively, and $\odot$ denotes element-wise multiplication. This module acts as a learnable band-pass filter, which can adaptively amplify attack-related frequency bands while suppressing random noise in background traffic. Z_{spec} is the final output spectrally enhanced feature sequence, which is then compressed into a vector $z_{spec} \in \mathbb{R}^{d_{model}}$ via max-pooling.

3.3.2. Discrete tower

The discrete tower focuses on processing the sequence of state codes of the protocol $X_{disc} \in \mathbb{R}^{L \times 1}$, with the aim of extracting the contextual logic of protocol interactions. Since the raw protocol state codes may be ambiguous, their specific meanings often depend on the physical attributes of the packets. Therefore, we innovatively inject the original physical features of the continuous tower as contextual information into the discrete embedding space.

We set the size of the protocol state code vocabulary to $|\mathcal{V}|$, which covers standard TCP Flags and custom UDP states. For the t th time step, the input embedding E_t of the discrete tower is calculated as follows:

$$E_t = Embed(x_t^{code}) + \mathbf{W}_{iat}(x_t^{iat}) + \mathbf{W}_{size}(|x_t^{size}|) + E_{pos}[t] \quad (18)$$

In Eq. (18), $Embed(\cdot)$ is the projection of the lookup table for the state codes of the protocol, mapping discrete integers to dense high-dimensional vectors; $\mathbf{W}_{iat}$ and $\mathbf{W}_{size}$ are linear projection matrices that map scalar physical features to the hidden layer; E_{pos} is a learnable positional embedding matrix. Since the Transformer architecture itself does not possess inductive biases for recurrence, we design this term to explicitly encode the relative position information of packets in the sequence, ensuring the model can correctly infer the strict temporal causal relationships of protocol interactions.

The embedding sequence with contextual information injected is fed into a multi-layer Transformer Encoder. Using its global self-attention mechanism, the model can capture long-range protocol dependencies and identify anomalous sequences that violate state machine logic. Finally, the discrete semantic feature vector $z_{disc} \in \mathbb{R}^{d_{model}}$ is obtained by max-pooling.

3.3.3. Auxiliary modules and feature fusion

To address the potential blind spots of the dual-tower structure in local spatial feature capture and global statistical distribution, we introduce two lightweight auxiliary modules and design a multi-source feature fusion mechanism. First, we design a multi-scale one-dimensional convolutional network to capture microscopic local attack fingerprints. The input $X_{cnn} \in \mathbb{R}^{L \times (2 + d_{model}/8)}$ is a concatenation of normalized continuous features and discrete features embedded shallowly, where the discrete features are compressed to dimensions $d_{model}/8$ to reduce computational cost. The specific formula is as follows:

$$H_{cnn} = Concat(Cony_{k_1}(X_{cnn}), Cony_{k_2}(X_{cnn}), Cony_{k_3}(X_{cnn})) \quad (19)$$

where the kernel sizes are set to $\{1, 3, 5\}$, respectively. This design enables it to simultaneously capture single-packet features, short-context interactions, and long-context patterns, with the output channel count

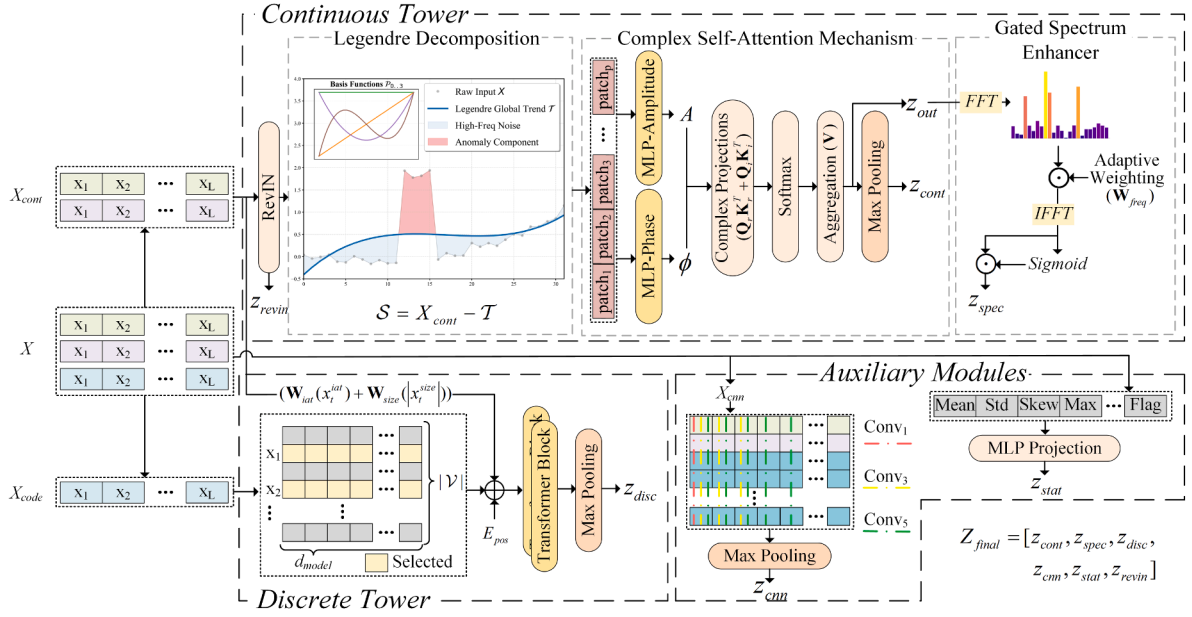


Fig. 2. Spectral-temporal dual-tower encoder.

unified with d_{model} . H_{cnn} is the concatenated result of multi-scale features, from which the local feature vector $z_{cnn} \in \mathbb{R}^{d_{model}}$ is obtained via max-pooling. Second, to preserve flow-level statistical characteristics, we compute the statistical feature vector z_{stat} , which covers the mean, variance, skewness of x_t^{lat} and x_t^{size} , as well as the proportion of specific protocol flags. A two-layer Multi-Layer Perceptron (MLP) is used to map low-dimensional statistics to a high-dimensional feature space, producing the statistical feature vector $z_{stat} \in \mathbb{R}^{2 \times d_{model}}$. Finally, we concatenate the feature vectors from the different views to construct the comprehensive flow representation Z_{final} :

$$Z_{final} = [z_{cont}, z_{spec}, z_{disc}, z_{cnn}, z_{stat}, z_{revin}] \quad (20)$$

In Eq. (20), $z_{revin} \in \mathbb{R}^4$ is the mean and variance of the continuous physical channel sequence extracted by the RevIN module. Although z_{stat} has already captured the high-order traffic distribution features, we also explicitly retain the feature parameters of the RevIN module. The motivation for this design is to alleviate the loss of physical magnitude information caused by normalization. The fused Z_{final} is the final embedding that contains multi-view information, which is used for subsequent multi-task pre-training and fine-tuning. The complete algorithm of the dual-tower encoder is shown in Algorithm 2.

3.4. Uncertainty-weighted multi-task pre-training

To enable the encoder to simultaneously learn robust representations of normal traffic and a keen perception of potential attack patterns, we design a multi-task pre-training scheme consisting of four sub-tasks. However, there are significant differences in the convergence speed and magnitude of different tasks, and simple linear weighting can easily lead to gradient dominance or optimization conflicts. Therefore, we introduce a dynamic weighting strategy based on homoscedastic uncertainty to adaptively adjust the weights of each task. The specific design is shown in Fig. 3.

The pre-training objective of PAST-IoT consists of four complementary tasks, which aim to constrain the model from four dimensions: feature space distribution, physical reconstruction, semantic understanding, and contextual inference.

- **Anomaly Contrastive:** To construct a compact manifold of normal traffic in the latent space, we adopt a margin-based pairwise distance loss. Given a benign sample input x_{pos} , we generate a corresponding augmented negative sample x_{neg} through the pattern-aware

Algorithm 2 Spectral-temporal dual-tower encoder.

Require: Continuous physical sequence $X_{cont} \in \mathbb{R}^{L \times 2}$, Discrete semantic sequence $X_{disc} \in \mathbb{R}^{L \times 1}$

Ensure: Fused comprehensive flow representation Z_{final}

1: // **Continuous Tower**

2: Apply RevIN on X_{cont} to handle non-stationarity, extract statistics z_{revin}

3: **Legendre Polynomial Trend Decomposition (Eq. 13):**

4: Compute global trend: $T = P \cdot W_{trf}(X_{cont})$

5: Isolate seasonal component: $S = X_{cont} - T$

6: **Complex Self-Attention Mechanism (Eq. 14, 15):**

7: Map S to complex domain to get $Z = A \cdot e^{j\phi}$

8: Compute complex score: $Score(Q, K) = \text{Re}(QK^H) = Q_r K_r^T + Q_i K_i^T$

9: Apply Softmax and aggregation to get complex sequence Z_{out}

10: **Gated Spectral Enhancer (Eq. 16, 17):**

11: Enhance frequency bands: $\tilde{Z}_{freq} = \text{IFFT}(\text{FFT}(Z_{out}) \odot W_{freq})$

12: Apply gating: $Z_{spec} = \text{Re}(\tilde{Z}_{freq}) \odot \text{Sigmoid}(W_{gate}(\text{Re}(\tilde{Z}_{freq}), \text{Im}(\tilde{Z}_{freq})))$

13: Max-pool output to obtain continuous feature z_{cont} and spectral feature z_{spec}

14: // **Discrete Tower**

15: **Context-Aware Embedding (Eq. 18):**

16: $E_t = \text{Embed}(x_t^{code}) + W_{lat}(x_t^{lat}) + W_{size}(|x_t^{size}|) + E_{pos}[t]$

17: Pass E through Transformer Encoder and apply max-pooling to obtain semantic feature z_{disc}

18: // **Auxiliary Modules & Fusion (Eq. 19, 20)**

19: Multi-scale CNN on shallow features: $H_{cnn} = \text{Concat}(\text{Conv}_1(X_{cnn}), \text{Conv}_3(X_{cnn}), \text{Conv}_5(X_{cnn}))$

20: Max-pool H_{cnn} to obtain local feature z_{cnn}

21: Extract high-order statistics and project via MLP to obtain z_{stat}

22: **Final Fusion:** $Z_{final} = [z_{cont}, z_{spec}, z_{disc}, z_{cnn}, z_{stat}, z_{revin}]$

23: **return** Z_{final}

anomaly injection mechanism (see Section 3.2 for details). We feed both into the encoder to obtain the latent representations z_{pos} and z_{neg} . The goal is to maximize the Euclidean distance between benign and anomalous samples, pushing it beyond a predefined margin m , thus forcing the model to learn discriminative features:

$$\mathcal{L}_{cont} = \frac{1}{B} \sum_{i=1}^B \max(0, m - \|z_{pos}^{(i)} - z_{neg}^{(i)}\|_2) \quad (21)$$

In Eq. (21), $\|\cdot\|_2$ denotes the L₂ norm, and B is the batch size. This loss function enforces that the distance between benign and anomalous samples is greater than the margin m .

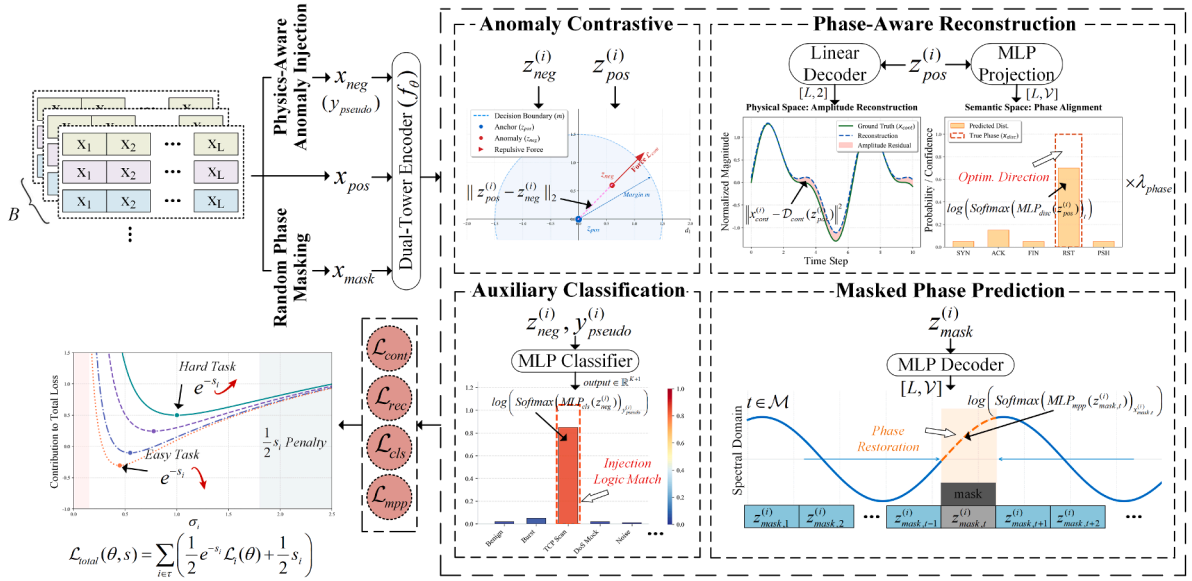


Fig. 3. Uncertainty-weighted multi-task pre-training.

lous samples in the feature space is at least m . This ensures that normal traffic forms a compact cluster in the latent space, while anomalous traffic is pushed away from this cluster, thereby learning the decision boundary for anomaly detection in an unsupervised manner.

- **Phase-Aware Reconstruction:** To ensure that the features retain the key physical and semantic information of the original traffic, we require the model to be able to reconstruct the original input from the latent representations. We introduce a phase-aware asymmetric reconstruction mechanism. Unlike the traditional autoencoder that optimizes a single unified loss, we decouple the decoding process into two orthogonal subspaces: the Physical Amplitude Space and the Semantic Phase Space. If only a single loss function is used, the gradients of large-magnitude continuous features often tend to obscure weak semantic signals. Therefore, we propose a dual-branch decoder structure with phase weighting:

$$\mathcal{L}_{amp} = \frac{1}{B} \sum_{i=1}^B \|x_{cont}^{(i)} - \mathcal{D}_{cont}(z_{pos}^{(i)})\|_2^2 \quad (22)$$

$$\mathcal{L}_{phase} = -\frac{1}{B} \sum_{i=1}^B \sum_{t=1}^T x_{disc,t}^{(i)} \log(\text{Softmax}(\text{MLP}_{disc}(z_{pos}^{(i)}))) \quad (23)$$

$$\mathcal{L}_{rec} = \mathcal{L}_{amp} + \lambda_{phase} \cdot \mathcal{L}_{phase} \quad (24)$$

Here, $x_{pos}^{(i)} = [x_{cont}^{(i)}, x_{disc}^{(i)}]$, and the linear decoder $\mathcal{D}_{cont}$ projects $z_{pos}^{(i)}$ back to the continuous feature space. To capture the complex protocol state transition logic, we adopt a non-linear projection head, a two-layer MLP with ReLU activation, which maps the latent vectors to the protocol space and computes the loss against the ground-truth labels after Softmax normalization. λ_{phase} is the phase emphasis factor, which explicitly amplifies the gradients in the phase space, forcing the encoder not only to preserve the energy information of the physical waveform but also to prioritize the correctness of the state logic of protocol when compressing features. In the figure, the model aligns the reconstructed waveform (blue dashed line) with the real physical features (green solid line) by minimizing the area of the pink shaded region. $\mathcal{L}_{phase}$ forces the prediction confidence (orange bars) to maximize the log-likelihood of the true protocol states (red dashed boxes), thereby ensuring precise logical decoding from the latent manifold.

- **Auxiliary Classification:** Simple contrastive learning only tells the model that samples are different, but does not specify how they dif-

fer. To leverage the prior knowledge generated during the anomaly injection phase, we introduce an auxiliary classification head after the encoder to perform fine-grained attack type recognition on samples. For each augmented input sample $x_{neg}^{(i)}$, the anomaly injection module synchronously generates a corresponding pseudo-label $y_{pseudo}^{(i)}$. We adopt a non-linear classification head MLP_{cls} to map the latent vector $z_{neg}^{(i)}$ to a K -dimensional attack category space. The auxiliary classification task aims to minimize the loss between the pseudo-labels and the predicted probability distribution:

$$\mathcal{L}_{cls} = -\frac{1}{B} \sum_{i=1}^B \log(\text{Softmax}(\text{MLP}_{cls}(z_{neg}^{(i)}))_{y_{pseudo}^{(i)}}) \quad (25)$$

By optimizing $\mathcal{L}_{cls}$, we endow the model with preliminary interpretability, making the learned feature space possess semantic clustering properties-i.e., similar attack patterns are closer to each other in the space.

- **Masked Phase Prediction(MPP):** The reconstruction task ensures the fidelity of local features, but it does not explicitly force the model to learn long-range temporal dependencies. Just as phase determines the current position of a wave within its cycle, protocol codes define the current stage of a communication session. Therefore, predicting the masked protocol codes is theoretically equivalent to reconstructing the missing phase information in the frequency domain, which is crucial for identifying stealthy attacks that manipulate protocol logic without producing significant anomalies in traffic volume. To this end, we propose MPP task and conceptually frame it as spectral recovery in the semantic domain.

For the discrete input sequence x_{disc} , we generate a corrupted view x_{mask} by replacing the tokens in a subset $\mathcal{M}$ with a learnable special token [mask]. $\mathcal{M}$ is obtained via random selection. The dual-tower encoder processes this corrupted sequence and outputs the corresponding latent representation sequence. For each masked time step $t \in \mathcal{M}$, we first extract its corresponding latent vector $z_{mask,t}$. This vector aggregates bidirectional contextual information through a self-attention mechanism. Subsequently, a two-layer MLP decoder is employed to perform spectral recovery directly in the latent space. The first layer projects $z_{mask,t}$ and is augmented with a GELU activation function to enhance the model's ability to parse non-linear phase patterns. The second layer acts as an output projection layer, mapping the features to the protocol vocabulary space $\mathbb{R}^{|V|}$, and finally generates the probability distribution of the missing token via

Softmax:

$$\mathcal{L}_{mpp} = -\frac{1}{B} \sum_{i=1}^B \frac{1}{|\mathcal{M}_i|} \sum_{t \in \mathcal{M}_i} \log \left(\text{Softmax}(\text{MLP}_{mpp}(z_{mask,t}^{(i)}))_{z_{mask,t}^{(i)}} \right) \quad (26)$$

By leveraging the non-linear mapping capability of the MLP, the model can more flexibly parse the complex phase patterns encoded in $z_{mask,t}$, thereby accurately reconstructing the protocol states at the physical level. In the figure, the orange dashed line intuitively shows the predicted output, visualizing the alignment accuracy between this recovered spectral phase curve and the ground-truth protocol states. Maximizing this term forces the model to generate a trajectory that can perfectly fill the gaps in the spectral manifold.

In the optimization process of multi-task learning, the core challenge lies in balancing the contributions of different tasks. In our framework, the loss functions of the four pre-training tasks exhibit significant heterogeneity in both physical meaning and numerical magnitude. Traditional linear weighting methods suffer from two critical theoretical flaws: first, the hyperparameter search space for weights is extremely large, leading to high computational costs; second, fixed scalar weights cannot adapt to the dynamic changes in the convergence speed of each task during training, which easily causes the model to be dominated by high-gradient tasks and fall into local optima. To address this optimization bottleneck, we adopt a homoscedastic uncertainty weighting strategy based on Bayesian deep learning, drawing inspiration from the foundational framework initially proposed for multi-task visual learning (Cipolla et al., 2018). Homoscedastic uncertainty describes the inherent observation noise of a task itself, which does not change with the input data but is learned by the model as an internal attribute of the task.

Based on the maximum likelihood estimation framework, we assume that the predictive output of the model for task i follows a Gaussian distribution (for regression tasks) or a Boltzmann distribution (for classification tasks) with a mean of f_θ and an observation noise variance of σ_i^2 . The goal of multi-task joint optimization is equivalent to minimizing the negative log-likelihood. To ensure numerical stability and avoid mathematical domain errors caused by negative variances, we use the log-variance $s_i = \log \sigma_i^2$ as a learnable parameter in the actual implementation. The final derived loss function is defined as follows:

$$\mathcal{L}_{total}(\theta, s) = \sum_{i \in \tau} \left(\frac{1}{2} e^{-s_i} \mathcal{L}_i(\theta) + \frac{1}{2} s_i \right) \quad (27)$$

In Eq. (27), the set of training tasks is $\tau = \{cont, rec, cls, mpp\}$, and θ denotes the set of weights for all shared encoders and task-specific decoders in the neural network. s_i quantifies the epistemic uncertainty of task i ; a larger value of s_i indicates that the model deems the current predictive noise of the task to be higher. As s_i increases, the weight e^{-s_i} decays exponentially. This enables the model to automatically suppress tasks with high noise or slow convergence, preventing their violently fluctuating gradients from damaging the shared feature space. The term $\frac{1}{2} s_i$ acts as a regularization constraint, which penalizes excessively high uncertainty and forces the model to find an optimal trade-off between reducing task weights and decreasing its own uncertainty.

The innovation and necessity of our uncertainty-weighted multi-task pre-training lie in three key aspects. First, unlike homogeneous visual tasks, our pre-training tasks span continuous physical waveforms and discrete protocol logic. By automatically inferring a larger s_i for high-gradient physical tasks, it exponentially decays their weight e^{-s_i} . This effectively prevents violently fluctuating gradients from overwhelming the discrete semantic inference, avoiding feature collapse. Second, s_i is not merely a numerical scaling factor; it acts as a dynamic indicator evaluating the coupling degree between physical transmission rhythms and semantic protocol states. As training proceeds and $\mathcal{L}_i(\theta)$ decreases, the model's confidence improves, smoothly driving s_i down to transition from coarse-grained learning to fine-grained tuning. Third, by automatically suppressing tasks with high observation noise during early pre-training, it prevents majority-class background traffic

from dominating the shared feature space. In this way, we achieve automatic balancing of heterogeneous tasks without manual intervention, ensuring the simultaneous learning of discriminative semantic features and high-fidelity physical features.

3.5. Fine-tuning

Considering the inherent extreme class imbalance of IoT network traffic data and the potential catastrophic forgetting problem of the model during fine-tuning, we design a fine-tuning strategy that integrates reconstruction regularization and dynamic sampling. To map high-dimensional manifold features to the specific attack category space, we construct a lightweight classification head. This module consists of a two-layer MLP, incorporating ReLU activation and Dropout regularization. Traditional fine-tuning methods only optimize the cross-entropy loss, which easily causes the model to overfocus on highly discriminative semantic features while discarding the underlying physical details critical for detecting unknown attacks. Therefore, we explicitly retain the phase-aware reconstruction loss as a regularization term during the fine-tuning phase. Second, to alleviate the model's overconfidence in mislabeled samples and enhance its generalization ability, we adopt the label-smoothed cross-entropy loss. The total fine-tuning loss is defined as:

$$\mathcal{L}_{fine} = \mathcal{L}_{fine-cls}(\hat{y}, y) + \mathcal{L}_{rec} \quad (28)$$

$$\mathcal{L}_{fine-cls}(\hat{y}, y) = -\frac{1}{B} \sum_{i=1}^B \sum_{k=1}^{K_{fine}} y_{i,k}^{LS} \log(\hat{y}_{i,k}) \quad (29)$$

$$y_{i,k}^{LS} = (1 - \gamma) \cdot y_{i,k} + \frac{\gamma}{K_{fine}} \quad (30)$$

In Eq. (28), $\hat{y}$ and y denote the predicted class probability distribution and the ground-truth class label, respectively; K_{fine} is the total number of classes in the downstream task; $y_{i,k}^{LS}$ represents the smoothed soft label target; and γ is the smoothing factor, which forces the model to maintain a moderate level of uncertainty at the decision boundary. In real-world IoT environments, benign traffic outnumbers attack traffic, and there exists a huge disparity in the sample size among different attack types. To address this, we design a weighted random sampling strategy to construct training batches, where the sampling weight of a class is inversely proportional to its occurrence frequency in the training set.

In summary, the aforementioned fine-tuning strategy constitutes the final critical step for the practical implementation of the PAST-IoT framework. Up to this point, the complete framework proposed in this paper, which encompasses pattern-aware anomaly injection, spectral dual-tower encoding, multi-task pre-training, and fine-tuning, has been fully elaborated. The next section will comprehensively verify the effectiveness and superiority of the framework through experiments and analysis.

4. Experimental results and analysis

To comprehensively evaluate the performance of the PAST-IoT framework on the task of IoT intrusion detection, extensive experiments are conducted on multiple real-world IoT traffic datasets in this chapter. We first introduce the datasets, evaluation metrics, and implementation details used in the experiments. Subsequently, we perform comparative experiments to benchmark PAST-IoT against state-of-the-art methods. Furthermore, we conduct ablation studies to deeply analyze the core contributions of the spectral-temporal dual-tower architecture, the pattern-aware anomaly injection mechanism, and the multi-task uncertainty weighting strategy. Finally, we present a visual analysis of its computational efficiency and the feature distribution in the latent space.

4.1. Experimental setup

To verify the model's ability to adapt to diverse network environments and attack patterns, we select four datasets, namely BoT-IoT

Table 2
Dataset details.

Dataset	Benign size	Attack category (size)
Pre-training dataset	312490	-
BoT-IoT	24750	Scan OS (2000), Service scan (2000), DDoS UDP (2000), DDoS TCP (2000), DDoS HTTP (2000), DoS UDP (2000), DoS TCP (2000), DoS HTTP (2000)
MQTT-IoT-IDS2020	25000	Mqtt Bruteforce (2000), Scan A (2000), Scan sU (2000), Sparta (2000)
IoT-Network-Intrusion	73968	SYN Flooding (2000), HTTP Flooding (2000), ACK Flooding (2000), Port Scanning (2000), Host Discovery (1509), UDP Flooding (500), Version Detection (422), ARP Spoofing (350), Telnet Bruteforce (163)
Edge-IIoTset	23149	Vulnerability Scanner (2000), Port Scanning (2000), SQL injection (2000), DDoS HTTP Flood (2000), Password (2000), DDoS TCP SYN Flood (2000), Uploading (2000), DDoS UDP Flood (2000), XSS (1225), Backdoor (848), Ransomware (614)

(Koroniotis et al., 2019), MQTT-IoT-IDS2020 (Hindy et al., 2020), IoT-Network-Intrusion (Kang et al., 2019), and Edge-IIoTset (Ferrag et al., 2022), for model fine-tuning evaluation. BoT-IoT is released by the Cyber Security Centre of the University of New South Wales, which mainly contains DDoS and DoS attacks launched by botnets. This dataset has an extremely high ratio of attack traffic to benign traffic, making it suitable for evaluating the model's anomaly detection performance under high throughput. MQTT-IoT-IDS2020 is the first dataset simulating an IoT network based on the MQTT protocol, including normal operations and 4 attack scenarios. It can effectively validate the model's ability to capture protocol semantic logic and subtle temporal features. IoT-Network-Intrusion is collected from a real IoT testbed with various heterogeneous devices. It includes man-in-the-middle attacks, RPL routing attacks, and other threats targeting specific device vulnerabilities. This dataset is adopted to verify the model's adaptability when facing highly heterogeneous device fingerprints. Edge-IIoTset is a modern dataset constructed for industrial IoT (IIoT) and edge computing scenarios. It covers not only traditional DDoS attacks but also injection attacks against industrial protocols such as Modbus, as well as multi-stage complex attacks including Backdoor and XSS. This dataset is used to evaluate the model's practicality in complex industrial control network environments. Detailed information is summarized in Table 2.

For the above four fine-tuning datasets, we randomly split each dataset into training, validation, and test sets with a ratio of 8:1:1. To simulate the label scarcity in real-world scenarios and validate the few-shot learning ability, during the fine-tuning stage, we select a certain proportion of training samples for model updating, and report the final performance metrics on the complete test set. For pre-training data, we use purely benign traffic, which covers various normal behavioral patterns such as device idle state, routine communication, firmware update, and user interaction.

In order to comprehensively evaluate the intrusion detection performance of PAST-IoT, we adopt the standard multi-dimensional evaluation metrics widely used in this field, including Accuracy, Precision, Recall, and F1-score. The standard F1-score is often biased by the majority class when processing imbalanced data. To more fairly evaluate the model's detection confidence and theoretical upper bound in complex backgrounds, we additionally introduce the following two metrics:

$$UAF1 = 2 \times \frac{\max\left(0, \frac{Precision - \lambda_{Anomaly}}{1 - \lambda_{Anomaly}}\right) \times Recall}{\max\left(0, \frac{Precision - \lambda_{Anomaly}}{1 - \lambda_{Anomaly}}\right) + Recall} \quad (31)$$

$$PA\%K = \frac{2 \sum_{i=1}^N y_i \cdot \mathbb{I}(o_i \geq \lambda_a)}{\sum_{i=1}^N y_i + \sum_{i=1}^N \mathbb{I}(o_i \geq \lambda_a)} \quad (32)$$

In Eq. (31), $\lambda_{Anomaly}$ represents the proportion of true anomaly samples in the test set of size N . y_i and o_i denote the true label and anomaly score of the i th sample, respectively. λ_a denotes the quantile threshold determined by the true proportion of anomalies. $\mathbb{I}(\cdot)$ denotes the indicator function, which outputs 1 if the condition in the parentheses holds, and 0 otherwise. UAF1 aims to eliminate the phenomenon of inflated precision

Table 3
Parameter configuration.

Parameters	Value	Parameters	Value
L	32	Complex Self-attention Layers	2
d_{model}	64	Transformer Layers	4
p	4	Attention Heads	4
D	3	Epochs-pre	30
m	1.5	Epochs-fine	20
λ_{phase}	2.0	lr-pre	1e-4
γ	0.1	lr-fine	5e-3
B	256	CNN Kernel	1, 3, 5
$ \mathcal{V} $	300	Masking Ratio(MPP)	0.5

caused by skewed class distribution, ensuring that the evaluation results reflect the model's true learning ability for attack features rather than statistical artifacts based on prior probabilities. PA%K evaluates the ranking quality of anomaly scores and the theoretical upper bound of the model's detection capability, verifying whether the model can correctly assign higher anomaly scores to attack samples than to benign samples.

The PAST-IoT framework is implemented based on the PyTorch 2.5.1 deep learning library, and is trained and tested on a high-performance server equipped with Ubuntu 20.04 LTS, an Intel(R) Xeon(R) Silver 4210R CPU, and an NVIDIA T4 GPU. The specific parameter configuration of the model architecture is designed to balance computational efficiency and feature representation capability. Detailed parameters are presented in Table 3.

4.2. Comparative experiments

To evaluate the performance of PAST-IoT, we conduct a detailed comparison with seven baseline models (FNN (Ullah & Mahmoud, 2022), IoT-ETEI (Yin et al., 2021), MTC-BYOL (Towhid & Shahriar, 2022), MTC-MAE (Xu et al., 2024), M2VT-IDS (Nie et al., 2024), DS-NIGNN (Pei et al., 2025), SSGMHAN (Diao et al., 2026)) on the above four datasets. FNN is a supervised anomaly detection model that uses a feedforward neural network to process flow features and flag features, representing the baseline performance of traditional deep learning in fully supervised scenarios. IoT-ETEI is an end-to-end IoT device identification and anomaly detection model that combines CNNs to extract features directly from raw traffic, representing existing end-to-end supervised learning schemes. MTC-BYOL is a contrastive learning-based encrypted traffic classification model that uses data augmentation to construct positive sample pairs, representing the main approach to contrastive self-supervised learning. MTC-MAE is a masked autoencoder-based self-supervised traffic classification model that learns traffic representations via random masking and reconstruction tasks, representing the application of self-supervised generative learning in the traffic domain. M2VT-IDS adopts a multi-view and multi-task learning architecture designed to handle multiple network tasks simultaneously, which serves as a strong baseline for multi-task learning architectures to compare the advantages of the uncertainty weighting strategy proposed in

Table 4

Comparison of intrusion detection performance of various models under samples with different proportion labels.

Method	1%				5%				10%			
	AC(%)	F1(%)	UAF1(%)	PA%K(%)	AC(%)	F1(%)	UAF1(%)	PA%K(%)	AC(%)	F1(%)	UAF1(%)	PA%K(%)
Bot-IoT												
FNN	85.42 ± 1.91	83.19 ± 1.28	93.96 ± 2.24	94.41 ± 1.93	90.94 ± 1.38	90.87 ± 1.17	98.57 ± 0.34	99.28 ± 0.15	94.20 ± 1.23	96.18 ± 0.96	99.34 ± 0.08	99.47 ± 0.09
IoT-ETEI	—	—	—	—	—	—	—	—	83.81 ± 3.46	77.55 ± 5.72	90.33 ± 1.69	92.63 ± 1.03
MTC-BYOL	81.63 ± 3.06	79.09 ± 3.12	92.35 ± 1.59	95.25 ± 1.05	89.17 ± 2.16	86.08 ± 3.43	99.24 ± 0.11	99.38 ± 0.14	91.83 ± 1.63	89.88 ± 1.27	99.21 ± 0.14	99.07 ± 0.05
MTC-MAE	83.21 ± 1.48	80.63 ± 1.93	95.06 ± 0.89	96.19 ± 0.74	95.19 ± 0.82	94.28 ± 0.77	99.18 ± 0.17	99.02 ± 0.19	99.12 ± 0.32	99.28 ± 0.21	99.15 ± 0.10	98.56 ± 0.13
M2VT-IDS	63.78 ± 4.63	50.21 ± 6.99	20.71 ± 8.09	82.65 ± 2.91	79.73 ± 3.86	77.42 ± 4.14	93.84 ± 2.08	97.73 ± 1.22	78.72 ± 1.93	79.23 ± 2.07	87.21 ± 2.88	95.83 ± 1.86
DS-NIGNN	64.36 ± 2.51	55.88 ± 3.07	68.79 ± 4.14	67.85 ± 3.95	75.17 ± 2.03	75.60 ± 1.72	83.49 ± 2.35	84.98 ± 2.01	80.83 ± 1.25	80.67 ± 1.14	90.59 ± 1.46	93.39 ± 1.05
SSGMHAN	71.28 ± 3.86	69.52 ± 3.71	44.09 ± 4.36	50.87 ± 3.55	75.93 ± 3.14	75.16 ± 3.01	52.08 ± 2.89	59.24 ± 2.75	87.30 ± 2.77	86.21 ± 2.72	77.64 ± 2.83	80.82 ± 2.37
PAST-IoT	85.59 ± 1.38	83.43 ± 1.45	94.14 ± 1.20	96.54 ± 1.01	95.82 ± 0.97	95.08 ± 0.90	99.43 ± 0.26	98.20 ± 0.47	99.26 ± 0.29	99.80 ± 0.12	99.52 ± 0.14	99.12 ± 0.19
IoT-Network-Intrusion												
FNN	93.13 ± 1.25	92.88 ± 1.18	71.12 ± 3.56	73.50 ± 3.21	94.88 ± 0.98	94.61 ± 0.83	78.31 ± 2.45	79.25 ± 2.18	94.89 ± 0.76	94.63 ± 0.72	78.33 ± 2.11	79.21 ± 1.98
IoT-ETEI	—	—	—	—	—	—	—	—	87.61 ± 2.12	76.44 ± 4.39	50.64 ± 9.23	61.96 ± 7.78
MTC-BYOL	83.08 ± 2.80	82.88 ± 2.76	84.24 ± 2.35	90.22 ± 1.89	92.24 ± 1.56	90.61 ± 1.78	90.85 ± 1.23	92.46 ± 1.15	95.88 ± 0.82	94.46 ± 0.91	91.27 ± 1.05	91.40 ± 0.92
MTC-MAE	91.24 ± 1.50	87.44 ± 1.83	93.58 ± 1.02	94.27 ± 0.98	95.05 ± 0.78	94.33 ± 0.81	94.38 ± 0.76	94.92 ± 0.69	95.15 ± 0.65	94.33 ± 0.71	95.71 ± 0.53	96.15 ± 0.48
M2VT-IDS	87.24 ± 3.21	81.47 ± 4.72	46.84 ± 6.58	44.59 ± 7.12	87.75 ± 2.89	63.68 ± 5.23	58.02 ± 5.89	59.87 ± 5.67	96.19 ± 1.05	96.73 ± 0.98	85.61 ± 3.12	87.33 ± 2.59
DS-NIGNN	85.56 ± 2.31	82.35 ± 2.19	31.84 ± 4.29	47.78 ± 3.98	89.26 ± 1.87	89.70 ± 2.01	69.89 ± 3.62	76.35 ± 2.48	91.18 ± 1.02	91.79 ± 0.85	76.53 ± 2.77	79.62 ± 2.49
SSGMHAN	76.25 ± 5.89	75.01 ± 5.14	22.52 ± 7.37	29.75 ± 6.40	76.90 ± 5.10	74.91 ± 4.99	50.32 ± 6.58	53.28 ± 7.19	78.82 ± 4.05	78.73 ± 3.96	72.26 ± 4.69	77.91 ± 5.55
PAST-IoT	94.10 ± 1.05	94.45 ± 0.98	90.23 ± 1.89	91.64 ± 1.76	96.40 ± 0.65	96.00 ± 0.72	92.05 ± 1.15	94.11 ± 0.89	96.25 ± 0.58	96.81 ± 0.61	92.07 ± 1.02	94.28 ± 0.78
Edge-IIoTset												
FNN	92.60 ± 1.12	92.40 ± 1.08	98.35 ± 0.56	99.44 ± 0.32	96.25 ± 0.89	96.30 ± 0.78	98.36 ± 0.45	98.57 ± 0.38	96.25 ± 0.76	96.23 ± 0.69	99.12 ± 0.28	99.53 ± 0.21
IoT-ETEI	86.59 ± 2.45	83.77 ± 2.89	99.21 ± 0.42	99.41 ± 0.35	95.25 ± 1.05	94.86 ± 0.98	99.35 ± 0.32	99.52 ± 0.28	98.15 ± 0.65	97.35 ± 0.58	99.07 ± 0.25	99.51 ± 0.19
MTC-BYOL	78.10 ± 3.25	72.49 ± 3.56	98.91 ± 0.48	99.28 ± 0.36	82.27 ± 2.89	77.74 ± 3.12	99.50 ± 0.28	99.38 ± 0.25	96.29 ± 0.78	95.04 ± 0.81	98.97 ± 0.32	99.05 ± 0.26
MTC-MAE	89.17 ± 1.89	84.97 ± 2.12	98.73 ± 0.52	98.79 ± 0.45	94.43 ± 1.02	92.97 ± 1.15	98.99 ± 0.38	99.14 ± 0.32	98.16 ± 0.62	97.95 ± 0.56	99.12 ± 0.26	99.26 ± 0.22
M2VT-IDS	—	—	—	—	70.06 ± 4.21	64.19 ± 4.89	96.35 ± 0.89	99.36 ± 0.42	84.59 ± 2.15	83.38 ± 2.46	99.19 ± 0.35	99.57 ± 0.28
DS-NIGNN	73.19 ± 1.96	66.51 ± 2.84	85.54 ± 1.59	85.19 ± 1.47	75.58 ± 1.15	74.20 ± 1.12	93.09 ± 0.97	92.51 ± 1.02	93.00 ± 0.64	92.35 ± 0.57	97.31 ± 0.66	98.37 ± 0.45
SSGMHAN	60.89 ± 6.61	54.48 ± 6.57	72.13 ± 5.05	73.73 ± 5.21	66.08 ± 5.29	61.05 ± 5.22	85.32 ± 2.63	87.41 ± 1.97	84.12 ± 2.16	81.86 ± 1.91	91.56 ± 1.09	94.28 ± 1.03
PAST-IoT	94.54 ± 0.98	94.19 ± 0.89	99.32 ± 0.35	99.36 ± 0.29	97.82 ± 0.76	97.48 ± 0.68	99.52 ± 0.25	99.62 ± 0.21	97.96 ± 0.59	97.57 ± 0.52	99.34 ± 0.22	99.62 ± 0.18
MQTT-IoT-IDS2020												
FNN	98.58 ± 0.85	98.55 ± 0.78	96.89 ± 1.02	97.12 ± 0.95	98.67 ± 0.65	98.64 ± 0.59	99.15 ± 0.42	98.25 ± 0.58	99.41 ± 0.48	99.38 ± 0.42	99.21 ± 0.35	99.08 ± 0.32
IoT-ETEI	—	—	—	—	—	—	—	—	83.81 ± 3.25	77.55 ± 4.89	59.29 ± 5.67	55.25 ± 6.12
MTC-BYOL	88.12 ± 3.18	84.06 ± 3.48	82.96 ± 4.21	95.87 ± 1.25	97.00 ± 1.02	96.64 ± 0.98	98.86 ± 0.56	98.94 ± 0.48	99.90 ± 0.25	99.89 ± 0.22	99.76 ± 0.18	99.81 ± 0.15
MTC-MAE	95.67 ± 1.25	95.33 ± 1.18	96.35 ± 0.98	97.75 ± 0.89	98.97 ± 0.76	98.81 ± 0.69	99.08 ± 0.45	99.18 ± 0.38	99.94 ± 0.22	99.92 ± 0.19	99.80 ± 0.16	99.83 ± 0.12
M2VT-IDS	—	—	—	—	74.49 ± 4.56	76.44 ± 4.23	80.66 ± 5.12	94.49 ± 1.89	78.34 ± 3.25	80.75 ± 3.18	83.72 ± 4.89	93.11 ± 2.15
DS-NIGNN	84.79 ± 3.04	79.44 ± 3.25	76.23 ± 4.31	74.39 ± 4.08	98.12 ± 0.61	98.20 ± 0.54	95.32 ± 1.03	98.85 ± 0.49	99.61 ± 0.11	99.60 ± 0.09	99.34 ± 0.14	98.85 ± 0.07
SSGMHAN	74.82 ± 4.86	64.25 ± 4.95	63.85 ± 6.88	72.50 ± 6.64	77.04 ± 3.60	76.56 ± 3.43	77.35 ± 4.69	78.76 ± 4.57	79.51 ± 3.15	71.43 ± 4.21	76.41 ± 4.11	79.90 ± 5.05
PAST-IoT	98.85 ± 0.78	98.87 ± 0.72	98.57 ± 0.85	97.79 ± 0.78	99.91 ± 0.02	99.93 ± 0.05	98.99 ± 0.38	98.76 ± 0.42	99.96 ± 0.02	99.92 ± 0.05	99.86 ± 0.10	99.90 ± 0.01

this paper. DS-NIGNN encodes the local network structure and traffic characteristics, representing a graph-based secure IoT intrusion detection system. SSGMHAN is a self-supervised graph multi-head attention network designed specifically for cloud-edge environments, representing the application of graph contrastive learning to capture spatial topological dependencies. The core of the experiments is to verify the effectiveness of the model in a few-shot learning scenarios. Therefore, we perform fine-tuning using labeled data in three different ratios: 1%, 5%, and 10%. The experimental results are presented in Table 4.

As can be seen in Table 4, PAST-IoT achieves the best overall performance on the four datasets. Especially in the few-shot scenario with 1% labels, PAST-IoT exhibits significant performance advantages. It is worth noting that graph-based architectures, due to their inherent topological compression, tend to eliminate fine-grained intra-flow dynamics when capturing spatial topologies. Their performance degrades significantly under the few-label scenario. On the Edge-IIoTset dataset, the F1-score of DS-NIGNN is merely 66.51%, whereas PAST-IoT achieves 94.19%. Similarly, in the MQTT-IoT-IDS2020 dataset, the F1-score of MTC-BYOL is only 84.06%, while PAST-IoT still maintains a high level of 98.87%. Such a considerable performance gap is mainly attributed to the pattern-aware anomaly injection mechanism of PAST-IoT. Different from the generic data augmentation used by MTC-BYOL, PAST-IoT enables the model to learn the intrinsic patterns of attacks in the pre-training stage by simulating physically meaningful anomalies such as Burst and DoS Mock. Therefore, even if the downstream task provides only a tiny number of labels, the model can quickly transfer the pre-trained features to the decision boundaries of specific attack categories.

In the IoT-Network-Intrusion dataset, the UAF1 and PA%K of MTC-MAE are slightly better than those of PAST-IoT. This is because MTC-MAE is a pure masked reconstruction-based model. When dealing with certain simple anomalies with only statistical significance, the single reconstruction error as the anomaly score is often highly sensitive, leading to better anomaly ranking and higher PA%K. However, such excessive sensitivity to statistical errors also results in blurred classification boundaries. Although PAST-IoT is slightly inferior in ranking metrics, it outperforms MTC-MAE by 2.86% in Accuracy and 7.01% in F1-score, which determine the final detection performance. This indicates that PAST-IoT achieves a reasonable trade-off in its design. Instead of merely pursuing the extreme value of reconstruction error, it considers both physical features and semantic logic simultaneously through the dual-tower architecture. At the cost of sacrificing a minor advantage in ranking, it gains more accurate discriminative ability for attack categories. This is more valuable in practical security scenarios, because misclassifying specific attack types is more critical than simply detecting the presence of anomalies. Under the 1% ratio on this dataset, although the Accuracy of FNN reaches 93.13%, its UAF1 drops sharply to 71.12%, which implies extremely weak recognition ability for minority-class attacks and heavy reliance on memorizing majority-class samples. In contrast, the UAF1 of PAST-IoT is as high as 90.23% and the PA%K reaches 91.64%. This further verifies the effectiveness of the spectral-temporal dual-tower encoder. Traffic generated by IoT devices usually contains complex periodic patterns. PAST-IoT leverages the complex self-attention mechanism to capture the coupling characteristics of phase and amplitude in the frequency domain. This enables it

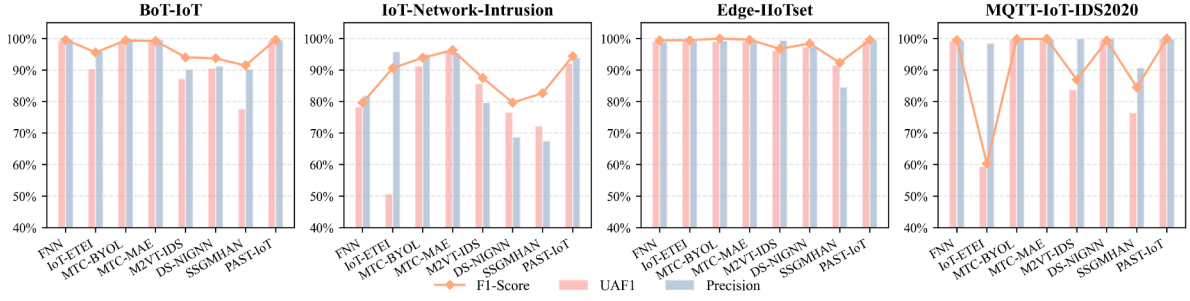


Fig. 4. Comparison of anomaly detection performance of different models under the 10% labeled data setting.

to accurately locate anomalies under heterogeneous noise backgrounds and maintain high sensitivity to long-tailed attacks even under extreme class imbalance.

To intuitively demonstrate the robustness and balance of the model for anomaly detection in label-scarce scenarios, Fig. 4 visualizes the multi-dimensional performance of each model under a 10% label ratio. It can be seen that PAST-IoT achieves the best and most balanced performance in all four datasets. Specifically, although FNN and DS-NIGNN achieve high precision on IoT-Network-Intrusion, their F1-score collapse severely. This indicates that although baseline models can identify obvious attacks well, they suffer from serious missing alarms and struggle to capture concealed anomalies. In contrast, PAST-IoT exploits the complex self-attention mechanism to extract phase-coupled information in the frequency domain, successfully filling this performance gap. Furthermore, on the MQTT-IoT-IDS2020 dataset, IoT-ETEI shows a drastic drop because it lacks the ability to explicitly model discrete protocol states, whereas PAST-IoT excels through its discrete semantic tower. Finally, although MTC-BYOL and MTC-MAE show strong competitiveness in Edge-IIoTset and BoT-IoT, the F1 curve of PAST-IoT still maintains a critical lead, and its UAF1 does not degrade due to data imbalance. This verifies that our uncertainty weighting strategy adaptively balances pre-training tasks, ensuring stronger discriminability and anti-interference ability.

4.3. Ablation experiments

To thoroughly analyze the contribution of each innovative component in the PAST-IoT framework to overall performance, we conduct a systematic ablation study based on the Edge-IIoTset and MQTT-IoT-IDS2020 datasets under the 1% label ratio setting. We divide the experiments into two parts: first, we evaluate the effectiveness of the model architecture and data augmentation mechanism; second, we investigate the influence of different pre-training task combinations on feature learning.

4.3.1. Effectiveness of architecture and data augmentation

To verify the necessity of the spectral-temporal dual-tower architecture and the pattern-aware anomaly injection mechanism, we construct the following four variants for comparison with the full model (PAST-IoT):

- **A1:** The Legendre polynomial trend decomposition module is removed, and the raw sequence is directly fed into the complex self-attention layer. This variant validates the contribution of separating global trends to capturing local subtle fluctuations.
- **A2:** The complex self-attention mechanism is replaced with the standard real-valued multi-head self-attention mechanism, which only processes amplitude information. This variant validates the role of frequency-domain phase information in detecting stealthy attacks.
- **A3:** The discrete tower is removed, with only the continuous tower retained. This variant aims to evaluate the importance of explicitly modeling protocol state interactions for understanding semantic attacks.

- **A4:** The pattern-aware anomaly injection mechanism is replaced with Gaussian noise and random masking. This variant validates the effectiveness of the negative sample generation strategy.

The experimental results are presented in Table 5. In addition, Fig. 5(a) intuitively visualizes the multi-dimensional performance distribution of each variant in terms of Precision, Recall, F1-score, and UAF1. On the Edge-IIoTset dataset, the performance profile of A4 exhibits the most significant inward collapse. Edge-IIoTset contains a large number of complex physical-layer attacks. Under the extreme condition of only 1% labels, the Gaussian noise and random masking adopted by A4 fail to simulate the temporal logic and physical patterns of these attacks. This result strongly demonstrates that pattern-aware injection, by introducing prior knowledge such as Burst and DoS Mock in the pre-training stage, serves as the cornerstone for the model to establish robust decision boundaries in few-shot scenarios. Without this mechanism, the model nearly loses its generalization ability when facing unseen physical attacks. On MQTT-IoT-IDS2020, the UAF1 of A3 drops to 89.81%, which is even lower than that of A4. This is mainly determined by the inherent characteristics of the dataset: MQTT attacks are often hidden in the contextual interactions of the protocol state machine, rather than in simple physical fluctuations of traffic. Removing the discrete tower means the model loses the ability to explicitly model protocol flags and status codes, making it difficult to capture logic-based protocol violations. This inversely verifies the necessity of the two-tower design in PAST-IoT. When A2 degenerates complex self-attention to real-valued self-attention, the model can no longer process frequency-domain phase information. This leads to a drop in its F1-score when identifying stealthy attacks in Edge-IIoTset that have subtle temporal amplitude variations but abnormal phases. This indicates that preserving the frequency-domain phase-coupled features of traffic is crucial for improving the granularity of detection.

4.3.2. Contribution of pre-training objectives

The core advantage of PAST-IoT lies in its multi-task pre-training strategy, which jointly optimizes four objectives: anomaly contrastive learning, phase-aware reconstruction, auxiliary classification, and masked phase prediction. To verify the contribution of each sub-task and the necessity of the uncertainty weighting strategy, we design the following four variants for comparison:

- **B1:** Only basic discriminative tasks are retained, i.e., the anomaly contrastive loss ($\mathcal{L}_{cont}$) and the auxiliary classification loss ($\mathcal{L}_{cls}$).
- **B2:** The phase-aware reconstruction task ($\mathcal{L}_{rec}$) is removed to evaluate the role of physical feature reconstruction in maintaining traffic fidelity.
- **B3:** The masked phase prediction task ($\mathcal{L}_{mpp}$) is removed to evaluate the contribution of contextual protocol inference to semantic understanding.
- **B4:** The uncertainty weighting strategy is removed, and all tasks are jointly optimized via a fixed-weight linear summation ($\mathcal{L}_{total} = \mathcal{L}_{cont} + \mathcal{L}_{rec} + \mathcal{L}_{cls} + \mathcal{L}_{mpp}$). This variant validates the key role of dynamic weight balancing in resolving multi-task gradient conflicts.

Table 5
Comparison of ablation experiment results.

Method	Edge-IIoTset				MQTT-IoT-IDS2020			
	PR(%)	RE(%)	F1(%)	UAF1(%)	PR(%)	RE(%)	F1(%)	UAF1(%)
A1	86.07 ± 1.25	83.35 ± 1.42	83.48 ± 1.38	93.22 ± 0.89	93.28 ± 0.89	92.02 ± 0.95	92.99 ± 0.92	95.90 ± 0.78
A2	85.69 ± 1.38	81.52 ± 1.56	84.67 ± 1.45	90.33 ± 1.02	94.47 ± 0.76	93.02 ± 0.81	93.74 ± 0.78	95.07 ± 0.85
A3	82.76 ± 1.89	79.28 ± 2.15	79.92 ± 2.08	85.31 ± 1.56	88.33 ± 1.25	85.72 ± 1.38	86.70 ± 1.32	89.81 ± 1.15
A4	64.53 ± 5.25	59.08 ± 7.56	60.09 ± 4.78	78.44 ± 3.12	91.08 ± 1.02	89.87 ± 1.08	90.88 ± 1.05	91.50 ± 0.98
PAST-IoT	94.97 ± 0.68	94.54 ± 0.72	94.19 ± 0.89	99.32 ± 0.35	98.93 ± 0.62	98.85 ± 0.58	98.87 ± 0.72	98.57 ± 0.85
B1	57.58 ± 3.56	49.21 ± 3.89	50.39 ± 3.78	60.77 ± 2.56	78.45 ± 1.56	73.80 ± 1.68	75.91 ± 1.62	88.64 ± 1.25
B2	82.71 ± 1.89	83.99 ± 1.76	83.28 ± 1.82	87.99 ± 1.25	89.26 ± 1.02	88.70 ± 1.08	88.86 ± 1.05	94.92 ± 0.89
B3	79.31 ± 2.12	75.18 ± 2.35	77.59 ± 2.28	85.44 ± 1.56	83.97 ± 1.25	80.88 ± 1.38	82.19 ± 1.32	90.52 ± 1.15
B4	84.44 ± 1.65	80.09 ± 1.78	81.65 ± 1.72	90.71 ± 1.08	92.17 ± 0.89	91.87 ± 0.95	92.04 ± 0.92	95.17 ± 0.85
PAST-IoT	94.97 ± 0.68	94.54 ± 0.72	94.19 ± 0.89	99.32 ± 0.35	98.93 ± 0.62	98.85 ± 0.58	98.87 ± 0.72	98.57 ± 0.85
C1	92.87 ± 0.49	90.31 ± 0.54	92.24 ± 0.50	99.05 ± 0.26	98.61 ± 0.80	97.55 ± 0.75	97.94 ± 0.79	97.82 ± 0.69
C2	83.75 ± 1.67	82.33 ± 1.84	83.49 ± 1.52	93.79 ± 0.92	84.28 ± 1.06	83.91 ± 0.91	83.99 ± 0.98	93.90 ± 0.71
C3	87.16 ± 0.90	87.04 ± 0.81	87.11 ± 0.84	95.76 ± 0.66	90.37 ± 1.11	89.66 ± 1.06	90.05 ± 0.97	97.01 ± 0.84
PAST-IoT	94.97 ± 0.68	94.54 ± 0.72	94.19 ± 0.89	99.32 ± 0.35	98.93 ± 0.62	98.85 ± 0.58	98.87 ± 0.72	98.57 ± 0.85

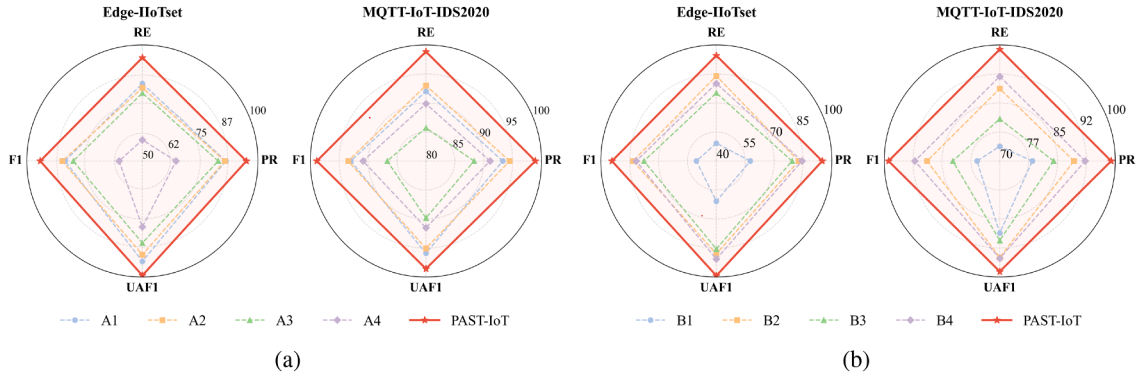


Fig. 5. Multidimensional performance comparison. (a) Performance comparison of ablation variants A1-A4; (b) Performance comparison of ablation variants B1-B4.

The experimental results are presented in Table 5. Fig. 5(b) further visualizes the performance coverage of each variant in different dimensions. In the highly heterogeneous Edge-IIoTset dataset, the F1-score of B4 is only 81.65%, a decrease of 12.54% compared with the full model. This significant gap reveals the gradient conflict problem in multi-task learning. Without the regulation of dynamic weights, high-gradient reconstruction tasks tend to dominate the optimization direction, forcing the encoder to overfocus on the restoration of physical waveforms at the expense of learning discriminative semantic features. PAST-IoT adaptively balances task weights using homoscedastic uncertainty, ensuring that the model achieves optimality in both physical fidelity and semantic discriminability, thereby avoiding the optimization process from falling into local minima. On the MQTT-IoT-IDS2020 dataset, the performance degradation of B3 is significantly higher than that of B2, which is highly consistent with the characteristics of the MQTT protocol. Attack behaviors in MQTT are mainly reflected in logical violations of the protocol state machine, rather than physical mutations of traffic payloads. The poor performance of B3 indicates that the absence of the MPP task deprives the model of the ability to infer missing protocol states in the discrete space, thus losing the capacity to understand long-range protocol context. Conversely, on Edge-IIoTset where physical features dominate, retaining the reconstruction task is more critical. This verifies the complementarity of the MPP and reconstruction tasks in semantic understanding and physical fitting, respectively. B1 exhibits the worst performance on both datasets, which fully demonstrates that relying solely on contrastive learning is insufficient to construct a sufficiently robust feature space in the few-shot scenario with only 1% labels. Reconstruction and prediction play a crucial regularization role: they force the model to understand the underlying generative mechanisms of the data, enabling it to extract generalizable representations even when discriminative tasks lack sufficient supervision signals.

4.3.3. Justification of algorithmic choices

To further verify the rationality and optimality of the core algorithm selection of PAST-IoT, we adopt alternative algorithms widely used in time series analysis and multi-task learning for comparative experiments:

- C1: Replace Legendre Polynomials with Moving Average (MA) for comparison.
- C2: Replace Complex Self-Attention with FFT + Real-valued MLP for comparison.
- C3: Replace Uncertainty Weighting with Dynamic Weight Averaging (DWA) for comparison.

The experimental results are presented in Table 5. For variant C1, where Legendre Polynomials are replaced with MA, the F1-score on the Edge-IIoTset dataset drops to 92.24%. This indicates that traditional filtering methods introduce inherent phase delays, whereas the global orthogonal fitting of Legendre Polynomials achieves zero-delay trend decoupling. Furthermore, variant C2, which substitutes Complex Self-Attention with the conventional FFT + Real-valued MLP, achieves an F1-score of only 83.49%. This sufficiently demonstrates that traditional real-valued networks completely sever the inherent coupling properties in frequency domains. By contrast, the complex-domain inner product adopted in our method plays an irreplaceable role in accurately capturing phase shift fingerprints of covert attacks. Finally, variant C3, which replaces Uncertainty Weighting with DWA, results in an F1-score of 87.11%. The reason is that DWA is incapable of balancing the substantial gradient discrepancies between heterogeneous physical and semantic tasks, which easily drives the model into local optima. The above ablation results strongly verify the rationality of the algorithm selection in our proposed framework.

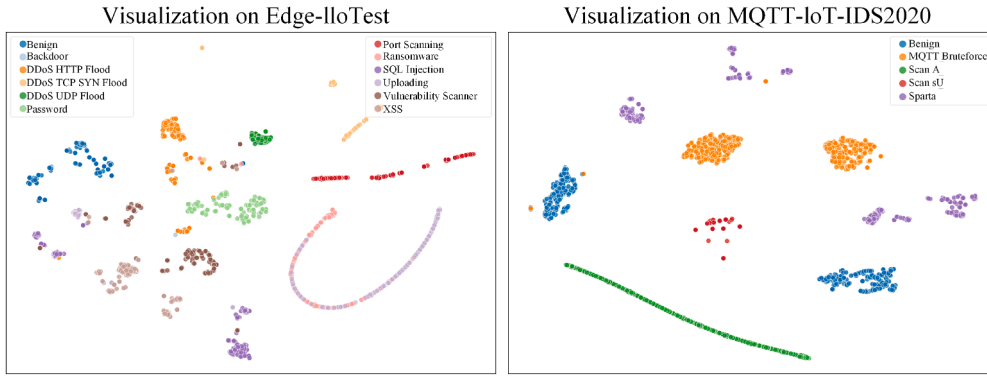


Fig. 6. Perform t-SNE visualization of the learned feature representations on Edge-IIoTset and MQTT-IoT-IDS2020.

4.4. Interpretability and visualization

To further verify whether the model has truly learned discriminative physical and semantic representations rather than merely fitting statistical biases, this section conducts a qualitative analysis from both macro and micro dimensions. First, we use t-SNE to visualize the high-dimensional feature space and evaluate the clustering ability of the model for different attack categories. Second, through a reconstruction case analysis, we demonstrate the model's deep understanding of the logic of covert attacks.

4.4.1. Embedding visualization

To intuitively evaluate the ability of PAST-IoT to construct manifolds in the latent feature space, we extract the final fused feature vectors of the model on the test set and project them onto a 2D plane using the t-SNE algorithm. Fig. 6 shows the feature distribution of the model on two datasets: the highly heterogeneous Edge-IIoTset and the protocol-specific MQTT-IoT-IDS2020, both using 5% labeled data.

In the Edge-IIoTset environment covering more than 10 attack types, PAST-IoT exhibits exceptional feature disentanglement capability. The model successfully maps physical-layer attacks based on traffic flooding and semantic-layer attacks based on payload content to distinct feature quadrants. This indicates that the two-tower spectral-temporal structure effectively decouples physical fluctuations and semantic logic, avoiding feature confusion between attacks of different patterns. For DDoS HTTP Flood and DDoS TCP SYN Flood, which have extremely similar physical features, the model still constructs clear decision boundaries. This is attributed to the precise encoding of protocol headers by the discrete tower, enabling the model to capture subtle protocol differences even under the same flooding attack type. Notably, Uploading and Ransomware exhibit a unique long-arc manifold in the feature space and are adjacent to each other. Unlike instantaneous attacks such as scanning, both Uploading and Ransomware involve continuous data transmission processes. The long-arc distribution accurately maps the dynamic evolution of attack traffic in terms of temporal persistence and payload accumulation, proving that the Legendre trend decomposition module, through effective feature disentanglement, eliminates the interference of trends on local features while still fully mapping the global temporal evolution of traffic in the latent space via reconstruction constraints. The spatial proximity of Ransomware and Uploading reflects the data exfiltration behavior often associated with modern ransomware. Both exhibit physical characteristics such as high throughput, large packet size, and high-entropy payloads. The model classifies them into the same latent neighborhood, indicating that it transcends simple label memorization.

The performance on MQTT-IoT-IDS2020 differs from the polymorphism of Edge-IIoTset, as this dataset focuses on protocol logic attacks. It can be seen that attacks such as MQTT Bruteforce and Sparta form highly dense and independent clusters. Such a highly compact cluster structure implies that the model has deeply understood the whitelist pattern

of normal MQTT communication during the pre-training stage through pattern-aware anomaly injection. Scan-type attacks such as Scan A and Scan sU are pushed away from the normal traffic manifold in the space, which also verifies the effectiveness of our MPP task. By learning the contextual transitions of protocol states, the model can keenly identify scanning sequences that do not conform to the standard state machine logic. In summary, the visualization results strongly demonstrate that PAST-IoT does not mechanically memorize data but successfully maps heterogeneous raw traffic into a metric space with clear semantics and well-defined boundaries.

4.4.2. Phase-domain reconstruction and semantic reasoning case study

In this section, a representative sample of UDP sequential scanning is selected for a case study. The sample consists of two stages: active scanning and communication truncation, aiming to verify whether the model can capture the physical evolution trend of traffic and the deep protocol interaction logic. The visualization results are shown in Fig. 7.

Fig. 7(left) presents the reconstruction performance of the model on the physical channel. For the packet-size sequence, the model accurately recovers the alternating fluctuation pattern of forward scanning packets and reverse ICMP response packets. This confirms that the Legendre projection in the continuous tower successfully preserves the directionality and periodicity of traffic. During the active period from $t = 0$ to $t = 13$, although real traffic contains high-frequency jitter, the reconstructed curve exhibits obvious smoothing and denoising properties while retaining only the main trend. At the traffic truncation point $t = 14$, the reconstructed waveform does not show a vertical stepwise reset to zero, but a certain degree of inertial tailing. Such a nonlinear transition indicates that the model does not perform simple numerical memorization, but constructs a continuous physical manifold by learning historical temporal dependencies. Fig. 7(right) uses a heatmap to illustrate the multi-step inference process of the model in the MPP task. During the active stage, the model shows high sensitivity to the protocol state machine of UDP scanning. At even time steps, the model predicts UDP-Seq with high confidence. This probability distribution strongly proves that the discrete tower has successfully internalized the interaction rules of network protocols, rather than merely recognizing individual packets. After truncation occurs, the prediction distribution of the model undergoes a rapid phase transition, with a significant rise in the probability of Padding/Other. This phenomenon demonstrates that the model can make logical inference of flow termination at the semantic level according to the traffic reset feature at the physical layer, achieving effective alignment between physical awareness and semantic reasoning.

4.4.3. Quantitative analysis of phase necessity

To more directly verify the necessity of complex-domain spectral modeling from a quantitative perspective, we conduct fine-grained performance comparisons across attack categories with distinct characteristics on the Edge-IIoTset (1% label ratio). We categorize attacks into

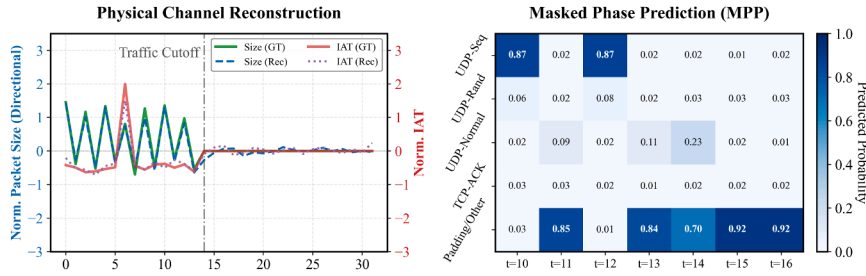


Fig. 7. Micro-case analysis based on UDP sequential scanning samples.

Table 6
Comparison results of fine-grained attack classification.

Attack Category	A2(AC(%))	A2(F1(%))	PAST-IoT(AC(%))	PAST-IoT(F1(%))
DDoS UDP Flood	96.34 ± 0.43	95.83 ± 0.36	96.91 ± 0.32	96.84 ± 0.30
DDoS TCP SYN Flood	94.05 ± 0.58	94.09 ± 0.46	95.19 ± 0.44	95.01 ± 0.37
Vulnerability Scanner	51.38 ± 2.87	49.14 ± 1.61	94.22 ± 0.28	95.04 ± 0.29
Port Scanning	57.64 ± 1.55	53.25 ± 1.27	92.82 ± 0.46	92.68 ± 0.38

two types: volumetric attacks (e.g., DDoS Flood), whose anomalies are mainly reflected in energy amplitude; and stealthy attacks (e.g., Port Scanning), whose traffic amplitudes highly overlap with normal background traffic. Their malicious characteristics are entirely concealed in the periodic misalignment of packet transmission timings, namely phase shifts in the frequency domain. We compare A2, which only perceives amplitude information while discarding phase information, with PAST-IoT. The experimental results are presented in Table 6.

For volumetric DDoS attacks, A2 and PAST-IoT achieve highly similar F1 scores. This indicates that amplitude features alone are sufficient for coarse-grained flood attack detection. Nevertheless, when confronting stealthy attacks, A2 suffers from catastrophic performance degradation due to its inability to capture temporal phase shifts in the frequency domain. Its F1 score for Vulnerability Scanner drops sharply from 95.04% to 49.14%. This striking contrast irrefutably demonstrates that the introduction of the complex-domain self-attention mechanism is indispensable for eliminating detection blind spots and accurately identifying stealthy attacks.

4.5. Parameter sensitivity and efficiency

In this section, we will deeply evaluate the robustness and operational efficiency of the PAST-IoT framework in the actual deployment environment. To determine the optimal configuration of the model and verify its adaptability to parameter changes, we conduct a sensitivity analysis of the key hyperparameters. Subsequently, considering the limited computing resources of IoT devices, we further evaluate the complexity of the model in detail.

4.5.1. Impact of key hyperparameters

To verify the robustness of PAST-IoT under different configurations and determine the optimal model structure, we conduct sensitivity analysis on four key hyperparameters on IoT-Network-Intrusion and Edge-IIoTset (with 5% labeled data), respectively: the input sequence length (L), the masking ratio for the MPP task, the degree of Legendre polynomials (D), and the feature embedding dimension (d_{model}). The experimental results are shown in Fig. 8.

As shown in Fig. 8, for the input sequence length (L), we set it within the range {8, 16, 32, 64}. On both datasets, the F1-score exhibits a trend of first increasing and then decreasing with the increase of length, reaching a peak at $L = 32$. When L is too short, the model struggles to capture the complete attack handshake cycle, leading to the loss of contextual information. Conversely, when L increases to 64, although the theoretical information volume increases, a large amount of invalid silent padding and background noise is actually introduced. For short-burst IoT attacks,

an overly long window dilutes the density of key features. Therefore, $L = 32$ is the optimal point that balances context capture and noise suppression. For the MPP masking ratio, the performance curve shows a significant inverted U-shape, achieving the optimal result at 0.5, and this pattern is highly consistent across both datasets. The masking ratio controls the learning difficulty of the discrete tower. A low masking ratio makes the reconstruction task too simple, allowing the model to complete interpolation using only neighboring flags without understanding the deep protocol interaction logic. An excessively high masking ratio, however, destroys too many contextual clues, making semantic reasoning infeasible. The degree of Legendre polynomials (D) determines the complexity of trend fitting. As D increases from 2 to 3, the model performance improves significantly, but starts to decline slowly when $D > 3$. When $D = 2$, the projection can only fit simple linear or quadratic trends, leading to underfitting and failing to effectively separate non-stationary noise. When $D = 3$, the model can accurately depict the polynomial envelope of traffic fluctuations, effectively assisting subsequent spectral feature extraction. However, as D further increases, higher-order polynomials begin to overfit local high-frequency jitters, interfering with the expression of global trends. This shows that $D = 3$ is the optimal balance between fitting capacity and smoothness. For the embedding dimension (d_{model}), performance improves substantially as it increases from 32 to 64; however, the performance gain saturates rapidly when increasing from 64 to 256. Although IoT traffic patterns are heterogeneous, their core features can be represented in a relatively compact vector space. $d_{model} = 64$ is sufficient to decouple the main attack patterns. Further increasing the dimension increases model capacity, but also significantly raises the number of parameters and inference latency, and may trigger overfitting risks in few-shot scenarios. Based on the principle of maximizing performance-efficiency, we select $d_{model} = 64$ as the final configuration.

4.5.2. Computational efficiency

To evaluate the deployment feasibility of PAST-IoT, we conduct a comparative analysis of its computational complexity and resource consumption with baseline methods under unified hardware configurations and the same batch size. The metrics adopted in this evaluation include: number of parameters (M), floating-point operations (FLOPs, G), memory footprint (MB), latency (ms), and throughput (samples/s). These metrics measure the model's spatial complexity, computational complexity, peak GPU memory usage during inference, average end-to-end latency for processing a batch of data, and the number of samples processed per second, respectively. Detailed experimental results are presented in Table 7.

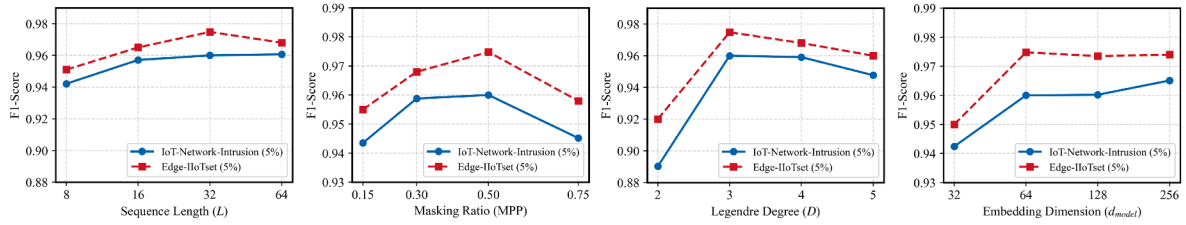


Fig. 8. Sensitivity analysis of key hyperparameters on IoT-network-intrusion and edge-IIoTset.

Table 7

Comparison of computational complexity among different models.

Method	Type	Params(M)	FLOPs(G)	Memory(MB)	Latency(ms)	Throughput(samples/s)
FNN	ML	4.5084	–	4.51	3.61	2410
IoT-ETEI	CNN + RNN	2.1251	0.0329	49.57	54.15	1211
MTC-BYOL	ResNet34	0.4251	0.0024	1.73	0.79	106547
MTC-MAE	Transformer	1.3320	0.1321	14.38	3.05	12830
M2VT-IDS	Transformer + LSTM	19.3165	5.0458	92.02	5.04	208
DS-NIGNN	GNN + Convolution	0.0442	0.0019	11.43	0.28	554374
SSGMHAN	GNN + Attention	0.0103	0.0048	64.74	0.49	22669
PAST-IoT	Dual-Tower	9.1168	0.0092	44.03	8.16	13881

As can be seen in Table 7, PAST-IoT exhibits exceptional efficiency and strikes an optimal balance between computational cost and detection capability. Despite introducing the complex dual-tower structure, the FLOPs of PAST-IoT are only 0.0092G, which is mainly attributed to the efficient feature disentanglement design avoiding redundant spatial computations. In terms of throughput, PAST-IoT reaches 13,881 samples/s, and its single-inference latency is only 8.16ms. This significantly outperforms the end-to-end model IoT-ETEI and the heavy multi-view model M2VT-IDS, making PAST-IoT easily meet the real-time requirements of high-bandwidth IoT gateways. It is worth noting that although DS-NIGNN, SSGMHAN, and MTC-BYOL exhibit extremely low parameter counts and latency, this extreme lightweighting comes at a high cost. Their F1 scores on multiple datasets have dropped significantly. In contrast, PAST-IoT has a moderate memory footprint, well within the limits of mainstream edge devices, while retaining high-resolution spectral-temporal features. In summary, PAST-IoT avoids the unacceptable accuracy loss of ultra-lightweight models and the massive overhead of heavy architectures, finding the optimal deployment sweet spot for complex IoT security scenarios.

5. Conclusions and future works

Faced with the severe challenges of the scarcity of high-quality labeled data and highly dynamic attack patterns in IoT environment, this paper proposes and systematically validates PAST-IoT, a pattern-aware self-supervised intrusion detection framework. It breaks the limitations of traditional intrusion detection models that excessively rely on manual annotations or single temporal statistical features. By introducing a pattern-aware anomaly injection strategy, we successfully endow the model with the capability to perceive the deep correlation between the physical attributes of network traffic and protocol specifications during the pre-training phase. Experimental results strongly demonstrate that the spectral-temporal dual-tower architecture constructed by PAST-IoT, leveraging Legendre polynomials and complex-valued self-attention mechanisms, can accurately decouple and capture the challenging periodic covert features in traffic. In addition, the multi-task learning mechanism based on homoscedastic uncertainty effectively alleviates the gradient competition problem in multi-objective optimization, ensuring the stability of the model under attack scenarios with varying complexity. Even in the extreme few-shot setting where only 1% labeled data is available, PAST-IoT not only maintains high robustness but also outperforms existing state-of-the-art models in multiple key metrics.

Although PAST-IoT achieves excellent performance, its anomaly injection mechanism is still limited by heuristic rules. Such manually

prior-guided injection approach imposes certain limitations on the theoretical upper bound of the model's defense against attacks with brand-new architectures. Our future work will focus on dynamic anomaly generation techniques. For example, generative adversarial networks (GANs) and conditional diffusion models will be utilized to adaptively generate hard adversarial samples according to the marginal distribution of real traffic. Meanwhile, we will also strive to develop lightweight online incremental learning algorithms to achieve full-life-cycle adaptive defense in IoT scenarios.

CRedit authorship contribution statement

Jiangtao Zhai: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization; **Lejun Shen:** Writing – review & editing, Methodology, Data curation, Conceptualization; **Shuo Wu:** Writing – review & editing, Supervision; **Kai Chen:** Writing – review & editing, Supervision; **Guangjie Liu:** Supervision, Resources.

Data availability

The implemented source code is in <https://github.com/IETAnalysis/PAST-IOT/tree/main>

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (Grants No. U2436601).

References

- Abdel Wahab, O. (2022). Intrusion detection in the IoT under data and concept drifts: Online deep learning approach. *IEEE Internet of Things Journal*, 9(20), 19706–19716. <https://doi.org/10.1109/JIOT.2022.3167005>
- Abusitta, A., de Carvalho, G. H. S., Wahab, O. A., Halabi, T., Fung, B. C. M., & Mamoori, S. A. (2023). Deep learning-enabled anomaly detection for IoT systems. *Internet of Things*, 21, 100656. <https://doi.org/10.1016/j.iot.2022.100656>
- Adil, M., Khan, M. K., Kumar, N., Attique, M., Farouk, A., Guizani, M., & Jin, Z. (2024). Healthcare internet of things: Security threats, challenges, and future research directions. *IEEE Internet of Things Journal*, 11(11), 19046–19069. <https://doi.org/10.1109/JIOT.2024.3360289>

- Alsulami, A., & Alturki, B. (2025). Enhancing multiclass network intrusion detection systems using continuous wavelet transform on network traffic. *Data & Metadata*, 4, 19. <https://doi.org/10.56294/dm2025474>
- Altaf, T., Wang, X., Ni, W., Yu, G., Liu, R. P., & Braun, R. (2023). A new concatenated multigraph neural network for IoT intrusion detection. *Internet of Things*, 22, 100818. <https://doi.org/https://doi.org/10.1016/j.iot.2023.100818>
- Cherfi, S., Boulaiche, A., Lemouari, A., & Abouaissa, A. (2026). Enhancing intrusion detection in IoT: CNN integration with K-means for efficient and balanced classification. *Expert Systems with Applications*, 299, 130122. <https://doi.org/10.1016/j.eswa.2025.130122>
- Cipolla, R., Gal, Y., & Kendall, A. (2018). Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In *2018 IEEE/CVF Conference on computer vision and pattern recognition* (pp. 7482–7491). <https://doi.org/10.1109/CVPR.2018.00781>
- Darban, Z. Z., Webb, G. I., Pan, S., Aggarwal, C. C., & Salehi, M. (2025). CARLA: Self-supervised contrastive representation learning for time series anomaly detection. *Pattern Recognition*, 157, 110874. <https://doi.org/10.1016/j.patcog.2024.110874>
- Diao, H., Li, X., Zhang, W., Yu, Y. S., Wang, Y., & Zhang, J. (2026). Self-supervised graph neural networks for network intrusion detection in cloud-edge collaboration environments. *Journal of Cloud Computing*, 15(29). <https://doi.org/10.1186/s13677-026-00847-z>
- Dillbary, N., Yozevitch, R., Dvir, A., Dubin, R., & Hajaj, C. (2024). Hidden in time, revealed in frequency: Spectral features and multiresolution analysis for encrypted internet traffic classification. In *2024 IEEE 21st consumer communications & networking conference (CCNC)* (pp. 266–271). <https://doi.org/10.1109/CCNC51664.2024.10454807>
- Eslami, E., & Hamouda, W. (2025). FedSSL-NTC: A robust federated self-supervised learning framework for network traffic classification under privacy constraints. *IEEE Open Journal of the Communications Society*, 6, 10023–10041. <https://doi.org/10.1109/OJCOMS.2025.3635689>
- Ferrag, M. A., Friha, O., Hamouda, D., Maglaras, L., & Janicke, H. (2022). Edge-IIoTset: A new comprehensive realistic cyber security dataset of IoT and IIoT applications for centralized and federated learning. *IEEE Access*, 10, 40281–40306. <https://doi.org/10.1109/ACCESS.2022.3165809>
- Gao, B., Liu, W., Liu, G., Nie, F., & Huang, J. (2025). Multi-level resource-cohered graph learning for website fingerprinting attacks. *IEEE Transactions on Information Forensics and Security*, 20, 693–708. <https://doi.org/10.1109/TIFS.2024.3520014>
- Huang, C., Wen, J., Xu, Y., Jiang, Q., Yang, J., Wang, Y., & Zhang, D. (2023). Self-supervised attentive generative adversarial networks for video anomaly detection. *IEEE Transactions on Neural Networks and Learning Systems*, 34(11), 9389–9403. <https://doi.org/10.1109/TNNLS.2022.3159538>
- H. Hindy, C. Tachtatzis, R. Atkinson, E. Bayne, X. Bellekens, MQTT-IoT-IDS2020: MQTT internet of things intrusion detection dataset <https://doi.org/10.21227/bhxy-ep04> (2020).
- H. Kang, D. H. Ahn, G. M. Lee, J. D. Yoo, K. H. Park, H. K. Kim, IoT network intrusion dataset <https://doi.org/10.21227/q70p-q449> (2019).
- Jadhav, D. S., Jadhav-Mane, S. R., Bamane, K. D., Mohite, S. S., Patankar, A. J., & Gaikwad, A. S. (2026). Intrusion detection in networks using ensemble-based deep federated learning. *Knowledge-Based Systems*, 339, 115491. <https://doi.org/10.1016/j.knsys.2026.115491>
- Koroniotis, N., Moustafa, N., Sitnikova, E., & Turnbull, B. (2019). Towards the development of realistic botnet dataset in the internet of things for network forensic analytics: Bot-IoT dataset. *Future Generation Computer Systems*, 100, 779–796. <https://doi.org/10.1016/j.future.2019.05.041>
- Li, R., Li, Q., Zhou, J., & Jiang, Y. (2022). ADRIoT: An edge-assisted anomaly detection framework against IoT-based network attacks. *IEEE Internet of Things Journal*, 9(13), 10576–10587. <https://doi.org/10.1109/IJOT.2021.3122148>
- Lin, K., Xu, X., & Gao, H. (2021). TSCRN: A novel classification scheme of encrypted traffic based on flow spatiotemporal features for efficient management of IIoT. *Computer Networks*, 190, 107974. <https://doi.org/10.1016/j.comnet.2021.107974>
- Lin, L., & Chen, Z. (2026). MPDetector: A multi-party collaborative federated transfer learning approach for IoT intrusion detection. *IEEE Transactions on Mobile Computing*, 25(1), 920–933. <https://doi.org/10.1109/TMC.2025.3600306>
- Liu, M., Jiang, B., Liu, J., & Li, K. (2026). Frequency-domain modulated spatio-temporal graph convolutional network for traffic flow prediction. *Expert Systems with Applications*, 312, 131474. <https://doi.org/10.1016/j.eswa.2026.131474>
- Liu, Z., Ma, Q., Ma, P., & Wang, L. (2023). Temporal-frequency Co-training for time series semi-supervised learning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(7), 8923–8931. <https://doi.org/10.1609/aaai.v37i7.26072>
- Luo, Y., Chen, X., Ge, N., Feng, W., & Lu, J. (2023). Transformer-based device-type identification in heterogeneous IoT traffic. *IEEE Internet of Things Journal*, 10(6), 5050–5062. <https://doi.org/10.1109/IJOT.2022.3221967>
- Luo, Y., Zhang, S., Lyu, Z., & Hu, Y. (2025). TFDNet: Time-frequency enhanced decomposed network for long-term time series forecasting. *Pattern Recognition*, 162, 111412. <https://doi.org/10.1016/j.patcog.2025.111412>
- Meidan, Y., Avraham, D., Libhaber, H., & Shabtai, A. (2023). CAdESH: Collaborative anomaly detection for smart homes. *IEEE Internet of Things Journal*, 10(10), 8514–8532. <https://doi.org/10.1109/IJOT.2022.3194813>
- Nandanwar, H., & Katarya, R. (2024). Deep learning enabled intrusion detection system for industrial IOT environment. *Expert Systems with Applications*, 249, 123808. <https://doi.org/10.1016/j.eswa.2024.123808>
- Nie, F., Liu, G., Liu, W., Huang, J., & Gao, B. (2025). IoT-AMLHP: Aligned multimodal learning of header-payload representations for resource-efficient malicious iot traffic classification. *Ad Hoc Networks*, 178, 103916. <https://doi.org/10.1016/j.adhoc.2025.103916>
- Nie, F., Liu, W., Liu, G., & Gao, B. (2024). M2VT-IDS: A multi-task multi-view learning architecture for designing IoT intrusion detection system. *Internet of Things*, 25, 101102. <https://doi.org/10.1016/j.iot.2024.101102>
- Pei, X., Deng, X., Tian, S., Jiang, P., Zhao, Y., & Xue, K. (2025). A privacy-preserving graph neural network for network intrusion detection. *IEEE Transactions on Dependable and Secure Computing*, 22(1), 740–756. <https://doi.org/10.1109/TDSC.2024.3417853>
- Ruan, T., Huang, Y., Zhu, Q., Hao, C., & Wu, Q. (2023). Multi-stage RF emitter search and geolocation with UAV: A cognitive learning-based method. *IEEE Transactions on Vehicular Technology*, 72(5), 6349–6362. <https://doi.org/10.1109/TVT.2023.3235498>
- Shen, M., Wu, J., Ye, K., Xu, K., Xiong, G., & Zhu, L. (2025). Robust detection of malicious encrypted traffic via contrastive learning. *IEEE Transactions on Information Forensics and Security*, 20, 4228–4242. <https://doi.org/10.1109/TIFS.2025.3560560>
- Thakkar, A., & Lohiya, R. (2023). Fusion of statistical importance for feature selection in deep neural network-based intrusion detection system. *Information Fusion*, 90, 353–363. <https://doi.org/10.1016/j.inffus.2022.09.026>
- Towhid, M. S., & Shahriar, N. (2022). Encrypted network traffic classification using self-supervised learning. In *2022 IEEE 8th international conference on network softwarization (netsoft)* (pp. 366–374). <https://doi.org/10.1109/NetSoft54395.2022.9844044>
- Tsogbaatar, E., Bhuyan, M. H., Taenaka, Y., Fall, D., Gonchigsumlaa, K., Elmroth, E., & Kadobayashi, Y. (2021). De-IoT: A deep ensemble learning approach to uncover anomalies in IoT. *Internet of Things*, 14, 100391. <https://doi.org/10.1016/j.iot.2021.100391>
- Ullah, F., Ullah, S., Srivastava, G., & Lin, J. C.-W. (2024). IDS-INT: Intrusion detection system using transformer-based transfer learning for imbalanced network traffic. *Digital Communications and Networks*, 10(1), 190–204. <https://doi.org/10.1016/j.dcan.2023.03.008>
- Ullah, I., & Mahmoud, Q. H. (2022). An anomaly detection model for IoT networks based on flow and flag features using a feed-forward neural network. In *2022 IEEE 19th annual consumer communications & networking conference (CCNC)* (pp. 363–368). <https://doi.org/10.1109/CCNC49033.2022.9700597>
- Wang, P., Song, Y., Xiang, Q., & Wang, X. (2026). Lifelong learning method for intrusion detection of internet of things with class-attention fusion strategy. *Knowledge-Based Systems*, 334, 115181. <https://doi.org/10.1016/j.knsys.2025.115181>
- Wang, S., Xu, W., & Liu, Y. (2023). Res-TranBiLSTM: An intelligent approach for intrusion detection in the internet of things. *Computer Networks*, 235, 109982. <https://doi.org/10.1016/j.comnet.2023.109982>
- Wang, Y., Xu, H., Ma, J., & Jin, Q. (2025). MTRC: A self-supervised network intrusion detection framework based on multiple transformers enabled data reconstruction with contrastive learning. *Journal of Network and Computer Applications*, 243, 104300. <https://doi.org/10.1016/j.jnca.2025.104300>
- Wu, J., & Wang, Y. (2026). TriHID: Towards verifiable domain adaptation-based IoT intrusion detection in heterogeneous environment. *Expert Systems with Applications*, 298, 129543. <https://doi.org/10.1016/j.eswa.2025.129543>
- Xiong, Y., Dong, S., Liu, R., Shi, F., & Jing, X. (2023). IoT network traffic classification: A deep learning method with fourier transform-assisted hyperparameter optimization. *Frontiers in Physics*, Volume 11–2023. <https://doi.org/10.3389/fphy.2023.1273862>
- Xu, K., Zhang, X., Wang, Y., Ohtsuki, T., Adebisi, B., Sari, H., & Gui, G. (2024). Self-supervised learning malware traffic classification based on masked autoencoder. *IEEE Internet of Things Journal*, 11(10), 17330–17340. <https://doi.org/10.1109/IJOT.2024.3357072>
- Yin, F., Yang, L., Wang, Y., & Dai, J. (2021). IoT ETEI: End-to-end IoT device identification method. In *2021 IEEE Conference on dependable and secure computing (DSC)* (pp. 1–8). <https://doi.org/10.1109/DSC49826.2021.9346251>
- Zha, C., Pan, H., Bai, B., Wu, J., & Zhang, R. (2026). FlowXpert: Context-aware flow embedding for enhanced traffic detection in IoT network. *IEEE Transactions on Mobile Computing*, (pp. 1–18). <https://doi.org/10.1109/TMC.2026.3651574>
- Zhan, M., Yang, J., Jia, D., & Fu, G. (2025). EAPT: An encrypted traffic classification model via adversarial pre-trained transformers. *Computer Networks*, 257, 110973. <https://doi.org/10.1016/j.comnet.2024.110973>
- Zhang, F., Wang, M., Zhang, W., & Wang, H. (2025). THATSN: Temporal hierarchical aggregation tree structure network for long-term time-series forecasting. *Information Sciences*, 692, 121659. <https://doi.org/10.1016/j.ins.2024.121659>
- Zhao, R., Zhan, M., Deng, X., Li, F., Wang, Y., Wang, Y., Gui, G., & Xue, Z. (2024). A novel self-supervised framework based on masked autoencoder for traffic classification. *IEEE/ACM Transactions on Networking*, 32(3), 2012–2025. <https://doi.org/10.1109/TNET.2023.3335253>
- Zhao, Z., Wang, Y., Lai, Y., Yu, S., & Shi, J. (2026). MUFFIN: A meta-knowledge decoupling-based approach to few-shot IoT traffic classification. *IEEE Transactions on Networking*, 34, 411–426. <https://doi.org/10.1109/TON.2025.3598870>
- Zhong, M., Zhong, T., Wang, Y., Chen, K., & Zhou, F. (2026). Noise resistant encrypted malicious traffic detection through kernel-enhanced contrastive view alignment. *IEEE Transactions on Networking*, 34, 1511–1523. <https://doi.org/10.1109/TON.2025.3625606>
- Zhu, S., Xu, X., Gao, H., & Xiao, F. (2023). CMTSNN: A deep learning model for multi-classification of abnormal and encrypted traffic of internet of things. *IEEE Internet of Things Journal*, 10(13), 11773–11791. <https://doi.org/10.1109/IJOT.2023.3244544>
- Zhuang, W., Fan, J., Fang, J., Fang, W., & Xia, M. (2024). Rethinking general time series analysis from a frequency domain perspective. *Knowledge-Based Systems*, 301, 112281. <https://doi.org/10.1016/j.knsys.2024.112281>
- Zong, Y., Aodha, O. M., & Hospedales, T. M. (2025). Self-supervised multimodal learning: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(7), 5299–5318. <https://doi.org/10.1109/TPAMI.2024.3429301>